

**Preparatory Commission for the Entry into Force of
the Agreement under the United Nations Convention on
the Law of the Sea on the Conservation and
Sustainable Use of Marine Biological Diversity of Areas
beyond National Jurisdiction and the Convening of the
First Meeting of the Conference of the Parties to the
Agreement**

Third session

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**Consolidated draft study on the technical aspects of the
operationalization of the Clearing-House Mechanism under
the Agreement under the United Nations Convention on the
Law of the Sea on the Conservation and Sustainable Use of
Marine Biological Diversity of Areas beyond National
Jurisdiction**

Background

1. The Agreement under the United Nations Convention on the Law of the Sea on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction (BBNJ Agreement) establishes a Clearing-House Mechanism (the Mechanism) and provides for its various functions. The specific modalities for the operation of the Mechanism will be determined by the Conference of the Parties to the Agreement.
2. In resolution 78/272 of 24 April 2024, the General Assembly established a Preparatory Commission to prepare for the entry into force of the BBNJ Agreement and the convening of the first meeting of the Conference of the Parties to the Agreement. The programme of work of the Commission is based on three clusters of issues discussed during the organizational meeting of the Commission, held in June 2024. One of these clusters addresses issues pertaining to the operation of the Mechanism, including the modalities for its operation, such as: (a) the type, architecture and functionalities of the platform; (b) the process for generating the “BBNJ” standardized batch identifier; (c) modalities to facilitate the matching of capacity-building needs with the support available and with providers for the transfer of marine technology, and to facilitate access to related know-how and expertise; and (d) terms of cooperation with relevant legal instruments and frameworks and relevant global, regional, subregional and sectoral bodies.
3. Pursuant to resolution 78/272, the Secretary-General of the United Nations, through the Office of Legal Affairs of the United Nations and the Division, is providing the Preparatory Commission with the assistance necessary for the performance of its work, including the provision of essential background information and relevant documents. In that context, the Preparatory Commission considered, at its first session, held in April 2025, issues pertaining to the operation of the Mechanism under the Agreement, on the basis of a note by the Secretariat ([A/AC.296/2025/6](#)).
4. With a view to providing essential background information to further support the work of the Commission with regard to issues pertaining to the Mechanism, the present study on the technical aspects of the operationalization of the Mechanism under the BBNJ Agreement (the Study) was prepared through a consultancy under the supervision of the Division.

5. The Study consists of four parts: (a) a stock-take study of existing clearing-house mechanism and information exchange systems, including safeguards for traditional knowledge of Indigenous Peoples and local communities, resources such as human resources, and cost-benefit analysis; (b) a compilation of options to address technical requirements for the Mechanism; (c) a draft roadmap for the phased operationalization of the Mechanism; and (d) a draft workplan for the initial phase for the operationalization of the Mechanism.

Part (a)

Stock-take study of existing clearing-house mechanisms and information exchange systems

I. Introduction

6. The stock-take study in Part (a) is based on desktop research and input provided by delegations to the Preparatory Commission, including through written submissions to inform the Study and interviews.

7. Section II of Part (a) provides a stock-take study of the existing clearing-house mechanisms and information exchange systems. There are two distinct categories of data, information, and documents that are relevant for the BBNJ Agreement. First, implementation of the BBNJ Agreement will generate data, information, and documents, therefore, the Mechanism would need to serve the function of a centralized platform that will “enable Parties to access, provide and disseminate information” of the four substantive elements (article 51, paragraph 3, subparagraphs (i) to (iv)). Second, the BBNJ Agreement requires that the Mechanism provide links to “relevant global, regional, subregional, national and sectoral clearing-house mechanisms and other gene banks, repositories and databases, including those pertaining to relevant traditional knowledge of Indigenous Peoples and local communities, and promote, where possible, links with publicly available private and non-governmental platforms for the exchange of information” (article 51, paragraph 3, subparagraph (c)). Thus, a hybrid of the so-called “centralized” and a “federated” model could be considered for the Mechanism, based on the two distinct but interrelated aspects of the needs associated with the implementation of the BBNJ Agreement. Section III provides considerations of relevant information sources for the four substantive elements.

8. Furthermore, the BBNJ Agreement provides that “full recognition shall be given to the special requirements of developing States Parties, as well as the special circumstances of small island developing States Parties, and their access to the mechanism shall be facilitated to enable those States to utilize it without undue obstacles or administrative burdens. Information shall be included on activities to promote information-sharing, awareness-raising and dissemination in and with those States, as well as to provide specific programmes for those States” (article 51, paragraph 5). To ensure that such requirement is considered, section III of Part (a) addresses the considerations relating to the special requirements of developing States Parties, as well as the special circumstances of small island developing States.

9. In relation to the remaining sections in Part (a), section IV explores examples of best practices associated with traditional knowledge of Indigenous Peoples and local communities; section V discusses examples of best practices on sensitive and confidential information, data sovereignty and cyber security; section VI explores considerations and different approaches to multilingualism; section VII provides cost-benefit analyses of two large and complex information exchange platforms; and section VIII outlines how this stock-take study would inform Part (b) of the Study.

II. Consideration of relevant information sources for the four substantive elements

10. This section considers the relevant information sources for the four substantive elements addressed by the BBNJ Agreement:

- (a) Marine genetic resources, including the fair and equitable sharing of benefits;
- (b) Measures such as area-based management tools, including marine protected areas;
- (c) Environmental impact assessments; and
- (d) Capacity-building and the transfer of marine technology.

11. The BBNJ Agreement requires the Mechanism to “foster enhanced transparency, including by facilitating the sharing of environmental baseline data and information” between Parties and other relevant stakeholders (article 51, paragraph 3, subparagraph (e)), and as discussed below, the same information source could be used for different purposes. Also, the requirement of the Agreement to “facilitate international cooperation and collaboration, including scientific and technical cooperation and collaboration” (article 51, paragraph 5, subparagraph (f)) is relevant for all four substantive elements. Therefore, the same existing information source may provide useful information for several elements. In addition, the useful nature of the information source is made on certain assumptions, because some of the details will be considered by the subsidiary bodies established under the Agreement (e.g., the Scientific and Technical Body, the Access and Benefit-Sharing committee, the Capacity-Building and Transfer of Marine Technology Committee, etc.) and decided by the Conference of the Parties at or after its first meeting. Such assumptions will be discussed where necessary for illustration purposes.

12. There is a wide range of sources for the types of information and data as discussed below. They are generated or held by intergovernmental bodies, regional bodies, sector-specific bodies, academic institutions, collections (e.g., natural history museums), non-governmental organizations and the private sector. It is important to note that there is also variability in the level of access.

A. Marine genetic resources, including the fair and equitable sharing of benefits

13. Regarding marine genetic resources of areas beyond national jurisdiction and digital sequence information on such resources, the BBNJ Agreement requires that Parties ensure that the Mechanism receives notifications prior to collection in situ of marine genetic resources of areas beyond national jurisdiction, upon which a “BBNJ” standardized batch identifier is automatically generated by the Mechanism. After collection in situ of marine genetic resource of areas beyond national jurisdiction, a post-collection notification must be submitted to the Mechanism, including information, along with the “BBNJ” standardized batch identifier, on the repository or database where digital sequence information on marine genetic resources was or would be deposited and where all marine genetic resources collected in situ were or would be deposited or held, among other information. It is worth noting that the functions of the Mechanism include receiving and maintaining the relevant notifications and generating the “BBNJ” standardized batch identifier, but do not including tracking and tracing marine genetic resources or digital sequence information.

14. In terms of repositories for physical samples of marine genetic resources, scientists use various biobanks and collection institutions. The Global Genome Biodiversity Network (GGBN) is a network of biodiversity repositories comprising 119 member repositories, with data providing members in the following States: Brazil, China, Colombia, Czechia, Ecuador, France, Germany, Hungary, India, Israel, Mexico, New Zealand, Peru, Poland, South Africa, Spain, Uganda, the United Kingdom of Great

Britain and Northern Ireland, and the United States of America. It aims to promote quality standards, improve practices, support interoperability and harmonize exchange of material in accordance with national legislations and international conventions. At the regional level, the European Marine Biological Resource Centre (EMBRC) connects institutions from nine countries. The Asian Network of Research Resource Centers (ANRRC) promotes cooperation between biobanks in China, Japan, the Republic of Korea and Southeast Asia. The International Seabed Authority (ISA) launched the Deep-Sea Biobank Initiative in June 2025 to enhance access to deep-sea biological samples and genetic data collected from the Area.

15. With regard to marine biodiversity data, scientists commonly use the Ocean Biodiversity Information System (OBIS), which is the world's largest open-access repository of marine biodiversity data. The system hosts various types of data, including taxonomic data, occurrence data, environmental data, metadata and biological details. OBIS operates globally, with thirty regional and seven thematic nodes, representing over 1,000 institutions and over 6,000 scientists and data managers from 99 countries. OBIS collaborates with numerous partners, including the Partnership for Observation of the Global Ocean (POGO), the Global Ocean Observation System (GOOS), the Global Biodiversity Information Facility (GBIF) and the Marine Biodiversity Observation Network (MBON), and contributes to initiatives such as the European Marine Observation and Data Network (EMODnet) and Pacific Islands Marine bioinvasions Alert Network (PacMAN).

16. For digital sequence information, the International Nucleotide Sequence Database Collaboration (INSDC) is a long-standing partnership among three primary databases that exchange data every 24 hours to ensure identical content:

(a) GenBank, managed by the National Center for Biotechnology Information of the United States of America;

(b) European Nucleotide Archive, managed by the European Bioinformatics Institute; and

(c) DNA Data Bank of Japan, managed by the National Institute of Genetics of Japan.

17. Approximately 95 per cent of other biological databases dealing with digital sequence information directly link to, or download their data from, the INSDC. These databases assign unique accession numbers to sequences. Other digital sequence information and genomic repositories include the China National GenBank and National Genomics Data Center of China, which collaborate with INSDC. INSDC is preparing to integrate the "BBNJ" standardized batch identifier into its existing infrastructure by utilizing free-text metadata fields where submitters currently enter cruise identifier. That said, INSDC would not enforce the requirement; responsibility for compliance would rest with the submitter. Interoperability with INSDC would benefit the Mechanism.

18. The BBNJ Agreement also requires notification to be submitted to the Mechanism where marine genetic resources of areas beyond national jurisdiction, and where practicable, the digital sequence information on such resources are subject to utilization, including commercialization. Such notification would need to include information, including the "BBNJ" standardized batch identifier, if available, on where the results of the utilization, such as publications, patents granted, if available and to the extent possible, and products developed, can be found, among other information. Examples of databases where information on utilized DNA may be found include UniProt (for protein sequence and functional information), as well as PubChem and ChEMBL (for small molecules and bioactive compounds). Data from three major patent offices, namely the United States Patent and Trademark Office, the European Patent Office and the Japan Patent Office, relating to patent applications containing biological sequences ultimately flow into INSDC.

19. OBIS is planning to pilot test a concept for the management of “BBNJ” standardized batch identifier with GenBank (an INSDC member). This pilot aims to demonstrate how such identifier could function technically by linking scientific data and digital object identifiers across systems.

B. Measures such as area-based management tools, including marine protected areas

20. The Agreement requires that the Mechanism serve as a centralized platform for accessing, providing and disseminating information on the establishment and implementation of area-based management tools, including marine protected areas. This may include supporting collaboration and consultations on the development of proposals for such areas, receiving proposals and making them publicly available, receiving input during consultations on such proposals and during consultations to enhance cooperation and coordination with relevant instruments, frameworks and bodies, making decisions of the Conference of the Parties and objections thereto publicly available, receiving and making publicly available reports and information on the implementation of area-based management tools, including marine protected areas, and providing general information on such tools.

21. One practical approach to conceptualize data related to measures such as area-based management tools, including marine protected areas is to consider three intersecting dimensions: first, the system to which the data relate (natural systems, human uses or governance arrangements); second, the form of the data (observed, modeled or derived, or administratively reported; and third, the level of access associated with the data (open, conditional or aggregated, or restricted). Any dataset relevant to the implementation of the BBNJ Agreement can be situated within this conceptual space, making explicit both its informational value and its limitations.

22. The use of such a data typology could help streamline the operationalization of the Mechanism with a more strategic focus on supporting decision-making. This approach may help clarify which types of data are most relevant at different stages of the processes relating to area-based management tools under the BBNJ Agreement, highlight where capacity-building efforts should prioritize analytical and integrative capabilities rather than raw data collection, and reinforce the idea that the primary function of the Mechanism is to support informed decision-making rather than to centralize control over ocean data. Such a conceptual framework may also help position the Mechanism as an effective interface between science, policy and governance, leveraging existing databases, information systems, and clearing-house mechanisms while respecting established mandates and sensitivities under the BBNJ Agreement. For the purpose of the present stock-take study, the first and third dimensions, namely the system to which the data relates and the level of access associated with the data, are explored.

23. For area-based management tools, including marine protected areas, the typology of data could be categorized in the three following groups:

(a) Natural systems: species occurrence and abundance; species distribution models, movement and ecological connectivity; genetic and genomic data, oceanography and bathymetry, and observed and modeled ecological data;

(b) Human uses and pressures: fisheries activities and effort; shipping and maritime traffic; seabed activities; submarine cables and infrastructure; observed and reported human-use data; and

(c) Governance and management: existing area-based management tools, including marine protected areas; spatial closures and regulations; maritime spaces limits and boundaries; institutional mandates; administrative and legal requirements.

24. Some widely used information systems and databases, such as the Ocean Biodiversity Information System (OBIS), the Global Biodiversity Framework Institute

(GBIF), AquaMaps, Movebank, the International Nucleotide Sequence Data Coalition (INSDC), the General Bathymetric Chart of the Oceans (GEBCO), the Global Ocean Observing System (GOOS), and the Copernicus Marine Service, provide globally, regionally and nationally relevant data and information via open access on species occurrences, modeled distributions, movement ecology, genetic diversity, and oceanographic conditions. The data and information they provide could support the establishment of ecological baselines, and the assessment of ecological representativity and connectivity. Such information and data may also support processes relating to the identification of areas for area-based management tools, as well as the establishment and monitoring of such tools.

25. Other sources of data relevant for area-based management tools, including marine protected areas, are held by States or intergovernmental bodies with sector-specific mandates, such as:

(a) Fisheries data collected by the Food and Agricultural Organizations of the United Nations (FAO) and regional fisheries management organizations, including systems such as the Vulnerable Marine Ecosystem (VME) Database, the Fisheries and Resources Monitoring System (FIRMS), and the Global Information Exchange System;¹

(b) Geological or environmental data generated through seabed exploration activities under the International Seabed Authority, including the Deep Seabed and Ocean Database (DeepData), and the Deep-sea Biobank Initiative; and

(c) Maritime regulatory information held by the International Maritime Organization (IMO), including the Global Integrated Shipping Information System (GISIS).

26. Access to fisheries data is subject to agreed criteria established by the providing State and FAO. Nevertheless, organizations such as the International Council for the Exploration of the Sea (ICES), the North Pacific Marine Science Organization (PICES) and the Pacific Community (SPC) use fisheries data released by FAO to provide open-access conservation analysis. The Convention for the Protection of the Marine Environment of the North-East Atlantic (OSPAR) uses data from its Parties to provide a wide range of information via open access through its OSPAR Data and Information Management System (ODIMS), including on biodiversity, area-based management, hazardous substances and eutrophication, radioactive substances, and human activities. For species and habitats, OSPAR supplements its own datasets with data from OBIS and GBIF to improve quality. While information held by the Commission for the Conservation of Antarctic Marine Living Resources (CCAMLR) could be informative, that commission indicated that, in the absence of a formal decision by its governing body regarding engagement with the BBNJ Agreement, it should be assumed that their publicly available information remains accessible only through the CCAMLR website and may therefore only be referenced.

27. Another category consists of data generated or curated by commercial or quasi-commercial entities, including satellite analytics providers and information systems such as the Global Fishing Watch, which offer high-temporal-resolution and near-real-time insights into human activities at sea, and automatic identification systems (AIS), which provide datasets containing shipping location data, destinations, and historical vessel positions. Some AIS-based information systems, such as the Marine Traffic and VesselFinder applications, provide limited free access, while more comprehensive information, including satellite-based tracking is available through paid subscriptions. Similarly, the IMO-Vega Database provides a comprehensive digital database of historical and current maritime regulations, but access is based on paid subscriptions. Access to more detailed datasets relevant to monitoring, compliance support and the

¹ FAO recommends that FIRMS serve as the primary source for fisheries-related data, given the data sovereignty and confidentiality considerations inherent to fisheries management.

evaluation of the effectiveness of area-based management tools, including marine protected areas, from commercial or quasi-commercial entities would likely require licensing agreements or partnerships.

28. The final category consists of a large volume of ecologically significant data and information held across academic institutions, museums, and individual research projects, including historical specimen records, data from research expeditions, telemetry studies, and unpublished or non-standardized datasets. These so-called “long-tail” data may be particularly important in data-scarce regions and for understudied taxa, such as deep-sea species, migratory megafauna and seamount-associated communities. In this context, the role of the Mechanism would be less about centralized data storage and more on improving discoverability, signaling where relevant information and data exist and under what conditions they may be accessed or reused. Data and information generated through the implementation of the BBNJ Agreement could help capture some of this information in a more standardized manner. As for existing “long-tail” data, the Scientific and Technical Body could conduct outreach to explore whether and how such data and information may be referenced in the Mechanism.

29. Table 1 below provides an illustrative overview of data and information sources relevant to area-based management tools, including marine protected areas, and reflects the requirement of the Agreement for the Mechanism to facilitate the sharing of environmental baseline data and information between Parties and other relevant stakeholders, as well as to facilitate international cooperation and collaboration, including scientific and technical cooperation and collaboration.

Table 1.

Illustrative data and information sources relevant to area-based management tools, including marine protected areas

<i>Data category</i>	<i>Data and information type</i>	<i>Illustrative sources</i>	<i>Primary use case</i>
Natural system	Species occurrence and abundance	OBIS, GBIF, FIRMS, WCPFC, NEAFC, SPC, ICES, PICES, OSPAR, World Ocean Database (WOD), GGBN	Ecological baselines; biodiversity hotspots
Natural system	Species distribution models and habitat suitability	OBIS, AquaMaps, FIRMS, OSPAR	Representativity; gap analysis; identification of areas for area-based management tools
Natural system	Movement and connectivity data	Movebank, Migratory Connectivity in the Ocean (MiCO), telemetry studies, WCPFC	Network design; migratory corridors
Natural system	Genetic and population structure data	INSDC, eBioAtlas	Defining management units; transboundary justification
Natural system	Abiotic and oceanographic data	GEBCO, Copernicus, GOOS, WOD, World Ocean Atlas	Environmental context; climate sensitivity
Human uses and pressures	Fisheries activity and effort	FIRMS, WCPFC, NEAFC, ICES, PICES, SPC, OSPAR, AIS, Global Fishing Watch, port authorities	Cumulative impacts
Human uses and pressures	Shipping and maritime traffic	AIS, WCPFC, Global Fishing Watch, port authorities,	Ship strikes; noise impacts
Human uses and pressures	Seabed activities and infrastructure	DeepData, Deep-Sea Biobank Initiative, TeleGeography	Scoping; cumulative impacts
Governance and management	Existing area-based management tools and spatial measures	MPAtlas, Protected Planet, VME Databases (FAO & regional)	Identifying gaps and overlaps
Governance and management	Legal and related information (e.g., meetings, documents, etc.) and maritime spaces limits and boundaries	BBNJ Agreement website and other UN DOALOS websites (such as the maritime space database), VME Databases (FAO & regional), InforMEA, IMO-Vega, Protected Seas	Jurisdictional analysis

C. Environmental impact assessment

30. The Agreement contains various provisions specifying the functions of the Mechanism in relation to environmental impact assessments, which include receiving and making available information and reports on assessments conducted under a Party's national process; publishing reports on assessments conducted under relevant instruments, frameworks and bodies; making relevant information on screening publicly available; making publicly available the views registered by Parties on the potential impacts of a planned activity and recommendations of the Scientific and Technical Body; publishing public notifications of planned activities; making draft assessment reports available during the public consultation process and publishing assessment reports; making decision documents on planned activities public; making monitoring reports public; publishing notifications of significant adverse impacts of authorized activities; making publicly available concerns registered by Parties that the authorized activity may have significant adverse impacts, any notifications issued and any recommendations by the Scientific and Technical Body; keeping States and stakeholders informed in the monitoring, reporting and review processes; and publishing reports on the review of the impacts of authorized activities and decision documents on changes to authorization decisions.

31. The typology of data discussed above could also be relevant for environmental impact assessments, subject to the advice of the Scientific and Technical Body, decision by the Conference of the Parties, and their implementation by Parties. While the sources of information and data may be the same or similar, the primary use thereof may differ. Area-based management tools, including marine protected areas, rely on such information and data to establish ecological baselines and identify biodiversity hotspots to determine whether a particular area in areas beyond national jurisdiction would require management or protection. Environmental impact assessments, in contrast, use similar information and data to determine, prior to a planned activity taking place, whether the planned activity may cause harm to the marine environment. In the latter context, environmental baseline information, such as species occurrence, ecological representativity and connectivity, and human-use data, would be valuable. The information sources identified in the section above as relevant to area-based management tools, including marine protected areas, may therefore also be relevant to environmental impact assessments.

32. In summary, table 2 below illustrates how selected types of data are used differently in the context of area-based management tools, including marine protected areas, and environmental impact assessments.

Table 2.

Illustrative differences in the primary use of selected data types for area-based management tools, including marine protected areas, and for environmental impact assessments

<i>Data category</i>	<i>Primary use for area-based management tools, including marine protected areas</i>	<i>Primary use for environmental impact assessment</i>
Species distribution models and habitat suitability products	Establish ecological baselines and identify biodiversity hotspots	Determine if sensitive species are in the direct area of a planned activity
Movement and connectivity	Support the design of ecologically coherent networks of area-based management tools, including marine protected areas	Assess whether sensitive species or ecological linkages may be impacted
Genetic and genomic data	Provide information on population structure, levels of gene flow and evolutionary connectivity (e.g., eDNA)	Assess the resilience and adaptive capacity of populations under environmental change
Abiotic and oceanographic data (including bathymetry, seafloor features and dynamic variables such as temperature)	Provide environmental context, support predictive modeling, and particularly important for anticipating climate-driven shifts in species distributions and ecosystem structure	Assess environmental context and cumulative impacts.
Human-use data	Identify activities to avoid or manage	Assess potential for substantial harm from the planned activity

33. Strategic environmental assessments are used to assess the potential effects of plans and programmes relating to activities in areas beyond national jurisdiction (article 39, paragraph 1). While the Conference of the Parties will develop guidance on the conduct of strategic environmental assessments, information similar to that used for environmental impact assessments may also be relevant for conducting strategic environmental assessments.

D. Capacity-building and the transfer of marine technology

34. The Mechanism is required to “facilitate the matching of capacity-building needs with the support available and with providers for the transfer of marine technology, including governmental, non-governmental or private entities interested in participating as donors in the transfer of marine technology, and facilitate access to related know-how and expertise” (article 51, paragraph 3, subparagraph (b)). The Mechanism may also facilitate the assessment of the needs and priorities of developing States Parties in terms of capacity-building and the transfer of marine technology.

35. The Intergovernmental Oceanographic Commission of UNESCO (IOC-UNESCO) has developed a clearing-house mechanism based on the generation, curation and sharing of data and information across several systems, including the Global Ocean Observation System (GOOS), the Ocean Biodiversity Information System (OBIS), the World Ocean Atlas, the World Ocean Database, the Ocean Teacher Global Academy (OTGA), the Ocean Capacity Development Hub (CD Hub) and Ocean Expert. These information systems are integrated and made interoperable through the Ocean Data and Information System (ODIS) and are accessible through the Ocean InfoHub, which serves as a user-facing interface to facilitate access, discovery and reuse of data and resources. Of particular relevance for capacity-building and the transfer of marine technology are OTGA, which provides online and in-person training opportunities to increase national and regional capacity in coastal and marine sciences, services and management; the CD Hub, which hosts information on a wide range of capacity-building opportunities, including training higher education programmes, research exchanges, mentorship and professional networks, as well as awards, fellowships and grants, among others. Ocean Expert, which enables users to identify experts, organizations and potential partners in ocean science and management. The information system is also used as an event management system for meetings and workshops, and currently hosts information on over 3,700 events and over 36,000 documents.

36. The United Nations Environment Programme (UNEP) is currently rebuilding the World Environment Situation Room, which will aggregate real-time geospatial data, earth observation data and environmental statistics. InforMEA aggregates decisions, treaty texts and national reports from over 20 multilateral environmental agreement secretariats. The UNEP-World Conservation Monitoring Centre enables users to search and filter datasets by theme. The Data Reporting Tool for Multilateral Environmental Agreements (DaRT) provides a workspace for national reporters to organize information for reporting across multiple biodiversity-related conventions simultaneously.

37. In addition to the IOC Ocean InfoHub and the UNEP information systems, table 3 below provides illustrative examples of existing databases and information systems complementing information that may be provided by developed States Parties and regional economic integration organizations under the Agreement.

Table 3.

Illustrative platforms relevant to capacity-building and the transfer of marine technology

<i>Data category</i>	<i>Subject of capacity-building and the transfer of marine technology</i>	<i>Illustrative sources</i>	<i>Primary use case</i>
Capacity-building and funding	Matters associated with the BBNJ Agreement	BBNJ Agreement website (https://www.un.org/bbnjagreement/en/capacity-building-and-technical-assistance/overview)	Open-access information system on the BBNJ Agreement, including information on the Agreement, official documents, workshops, technical assistance, briefings, as well as various tools and materials for capacity-building and awareness-raising
Capacity-building	Information on the Intergovernmental Conference on Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ IGC)	BBNJ IGC website (UN DOALOS) (https://www.un.org/bbnj/)	Open-access information system which provides historic information on the BBNJ IGC that negotiated and adopted the Agreement
Capacity-building, trust funds and fellowships	Information on the United Nations Convention on the Law of the Sea and the related agreements, as well as UN DOALOS activities	Oceans and Law of the Sea website (UN DOALOS) (https://www.un.org/depts/los/)	Open-access information system providing information on the United Nations Convention on the Law of the Sea and the related agreements, the work of the United Nations General Assembly on oceans and the law of the sea, bodies established by the Convention, settlement of disputes, the functions and activities of UN DOALOS, and calendar of meetings
Capacity-building and trust funds	Information on capacity-building and trust funds	UN DOALOS capacity-building website (UN DOALOS) (https://www.un.org/oceancapacity/)	Open-access information system providing consolidated information on technical cooperation, fellowships, advisory services, studies and guidelines, reference collection and databases, as well as trust funds administered by UN DOALOS, including information on funding support for developing State representatives to participate in the BBNJ Preparatory Commission meetings (BBNJ Trust Fund)
Capacity-building	Information on inter-agency coordination on marine and coastal issues through UN-Oceans	UN-Oceans website (UN DOALOS) (https://unsceb.org/un-oceans)	Open-access information system providing information on the mandates and work of UN-Oceans, including side events
Capacity-building support through experts	Support for ratification and implementation of the BBNJ Agreement	Global Ocean Programme established by the European Union and Technical Assistance Facility Component 1 implemented by the International Union for the Conservation of Nature (IUCN) (https://iucn.org/our-work/oceans-and-coasts/enhancing-capacities-high-seas-conservation)	Connects States and regional intergovernmental organizations with high-level expertise through coordinated, needs-based and targeted support from a list of approved experts
Capacity-building	Information to improve tuna and deep-sea fisheries management by strengthening regulatory frameworks and reducing their environmental impact	Common Oceans Program Knowledge Hub (FAO, United Nations Development Programme (UNDP) and UNEP) (https://www.fao.org/in-action/commonoceans/knowledge-hub/en)	Open-access information system providing information on regional and national training programmes for exchanging experiences and strengthening cross-sectoral collaboration
Capacity-building	Information and knowledge	IW: LEARN (funded by Global	Open-access information system

<i>Data category</i>	<i>Subject of capacity-building and the transfer of marine technology</i>	<i>Illustrative sources</i>	<i>Primary use case</i>
	management to support capacity development under the Global Environment Facility (GEF) International Waters focal area	Environment Facility, implemented by UNDP and UNEP and managed by IOC-UNESCO) (https://iwlearn.net/)	providing information resources and project information; open access
Funding	Ratification support, enabling activity guidelines and project database	GEF website on BBNJ (https://www.thegef.org/what-we-do/topics/international-waters/bbnj)	Open-access system providing information on funding for ratification support
Funding and matching	Marine research funding opportunities	Ocean Matcher (https://www.oceanmatcher.com/)	Facilitates the matching of submissions by donors and recipients with the support of artificial intelligence and natural language processing
Capacity-building and matching	Partnership opportunities for marine conservation that can be matched with partners	Ocean Connector (IOC-UNESCO) (https://www.oceanmatcher.com/)	Connects individuals and institutions needing capacity-building with those offering services, focusing on human skills, training opportunities, workshops, and technical mentorship, through staff manually reviewing and matching requests and partners
Capacity-building	Data and information to support scientific research	GBIF (https://www.gbif.org/)	Open-access compendium of some information and data, including studies using GBIF data; requires that users include attribution of access to GBIF; the GBIF Secretariat conducts literature tracking, which identifies research uses and citations of biodiversity information
Capacity-building	Development of national legislation, including considerations on enforcement.	SHERLOC (UN Office of Drugs and Crime (UNODC)) (https://www.unodc.org/cld/st/home.html)	Global knowledge management portal sharing electronic resources and laws on crime, including national jurisprudence, legal frameworks, open-access to most databases
Capacity-building	Law enforcement training	SPARQL / SPARK (UNODC) (https://elearningunodc.org/)	Online training modules with different tiers of access, including open-access training modules on anti-corruption, human rights, and organized crime, and restricted access to certain specialized training modules
Needs Assessment and matching on transfer of marine technology	Maritime technical cooperation	Technical Operations Portal (IMO) (https://webaccounts.imo.org/Common/WebLogin.aspx)	Functional but manual model for the “needs assessment and matchmaking on marine technology transfer”, with States submitting specific capacity-building needs which are then manually matched with funding sources and donors

III. Considerations relating to the special requirements of developing States Parties, as well as the special circumstances of small island developing States

38. Several information systems have taken measures aimed at making their systems remain accessible in low-bandwidth environments. For example, The Pacific Islands Protected Areas Portal (PIPAP) and the Pacific Community (SPC) Pacific Data Hub (PDH) utilize "light" web architecture to mitigate low-bandwidth challenges. The Pacific Ocean Portal offers three options to view the base map (i.e. OpenTopoMap, Satellite,

and SPC OSM) and specifically indicates that the SPC OSM basemap is recommended for high performance and faster loading under low-bandwidth conditions. The OSPAR Data and Information Management System (ODIMS) prioritizes stability and functionality through largely text- and HTML-based design to avoid the loading of heavy graphical elements. It was engineered with low computational power requirements and operates on a baseline configuration that expands automatically as demand increases. The platform used by the Ocean Biodiversity Information System (OBIS) also adopts a minimal design to support usability in low-bandwidth conditions.

39. Some information systems and clearing-house mechanisms have also developed mechanisms and processes to support offline accessibility. For example, the Global Biodiversity Information Facility (GBIF) initially operated as a federated network requiring real-time queries across multiple nodes. As this approach proved vulnerable to connectivity interruptions, GBIF later adopted a centralized indexing system and the use of dataset packages that allow users to perform a single download operation, often with high compression. Once the data are downloaded, they can be used entirely offline. Similarly, the Clearing-House Mechanism under the Convention on Biological Diversity (CBD) allows offline submissions via Excel or Word documents which can subsequently be uploaded by secretariat staff or helpdesk staff. The Biosafety Clearing-House under the Cartagena Protocol on Biosafety to the Convention on Biological Diversity also offers offline data-entry forms.

40. Addressing human resource constraints may be done in various ways. For example, integrating information systems such as InforMEA would allow users to search simultaneously across multiple multilateral environmental agreements without accessing separate systems. Stakeholders contacted for the purpose of the present Study noted that responsive user support, such as a helpdesk providing timely and meaningful support, can significantly improve usability, particularly for Parties with limited resources. The IMO Technical Operations Portal offers a functional, albeit largely manual, example of needs assessment regarding marine technology transfer. The use of tools such as the Data Reporting Tool for Multilateral Environmental Agreements (DaRT) of UNEP could also help reduce reporting burdens for small island developing States. More broadly, intuitive and easy-to-use platforms can help accommodate diverse users. The UNEP Global Partnership on Marine Litter platform, which requires user registration, provides an example of a system designed to serve a wide range of stakeholders.

41. Various information systems and clearing-house mechanisms also provide targeted support for small island developing States. For example, the Access and Benefit-Sharing Clearing-House under the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity includes modules that assist small island developing States in developing legislative frameworks for access and benefit-sharing. OBIS ensures that the Pacific-specific nodes provide data relevant to island scales (i.e., data is collected and analyzed and is aggregated to represent an entire island and its immediate surround coastal waters) so that it will be most useful for local decision-making. The Ocean Teacher Global Academy (OTGA) provides remote training opportunities that help overcome the geographic isolation faced by many small island developing States. The Pacific Islands Protected Areas Portal (PIPAP) was designed specifically to provide small island developing States with data and information about protected areas in the Pacific Islands, decision making support tools, case studies and how the Secretariat of the Pacific Regional Environment Programme can assist in meeting global biodiversity targets. The Pacific Climate Change Portal assists small island developing States in navigating Green Climate Fund procedures by consolidating relevant data and information. The Santiago Network for Loss and Damage also prioritizes requests from small island developing States for technical assistance.

IV. Examples of best practices associated with traditional knowledge of Indigenous Peoples and local communities

42. The Pacific Community (SPC) Pacific Data Hub (PDH) provides one of the most robust applications of the Local Contexts framework, which is a global initiative and software platform designed to address the unique intellectual property and data governance needs of Indigenous Peoples. Within SPC PDH, the use of Local Contexts is standardized as part of the regional mandate and is supported by the Pacific Data Sovereignty Network. The system is integrated into the regional data ecosystem to support Pacific data sovereignty. Sensitive traditional knowledge may be uploaded but digitally restricted through tags using the Local Contexts framework. Instead of a binary “public or private” classification, datasets may remain visible while access is restricted through cultural protocols. For example, a dataset may be tagged as “TK Women Restricted”. This approach allows the existence of data to be known while limiting access to authorized community members. The Global Biodiversity Information Facility (GBIF) is piloting an initiative where Indigenous Peoples apply the “TK and Biocultural Labels” developed under the Local Contexts framework. These labels enable communities to express conditions for the sharing and engagement with data and function as digital provenance metadata that maintain the connection to community interest as data move across repositories, thereby helping prevent the disclosure of sensitive content. GBIF is currently testing an application programme interface (API) to integrate these labels from the Local Contexts Hub into the GBIF portal.

43. The Access and Benefit-Sharing Clearing-House (ABSCH) under the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity exempts sensitive traditional knowledge from being uploaded. The internationally recognized certificate of compliance confirms that permission has been granted, but the traditional knowledge itself is not disclosed. In addition, various process-oriented safeguards for traditional knowledge may be incorporated into the procedures for submitting information to a clearing-house mechanism. For example, in Australia, traditional knowledge is managed through permits or contractual arrangements that are not publicly available, and public disclosure is permitted only where explicit consent has been provided. France employs a two-step process where initial notifications are submitted through a national online platform, while the verification of traditional knowledge involves on-site consultation with relevant communities. Reporting to the ABSCH occurs only after community consent and national authorization have been secured.

44. GBIF has also established an Indigenous Data Governance Task Group to guide the implementation of the Collective Benefit, Authority to Control, Responsibility and Ethics (CARE) Principles. The Convention for the Protection of the Marine Environment of the North-East Atlantic (OSPAR) has similarly incorporated the CARE Principles in initiatives relating to traditional knowledge, particularly in its work in the Arctic, ensuring that data provenance and licensing respect the interests of data originators.

45. The Observatory on Principle 10 in Latin America and the Caribbean (The Escazú Agreement Observatory) has a clearing-house and central tracking mechanism on environmental rights and laws, which provides information in Indigenous languages.

46. There is increasing recognition of the importance of traditional knowledge of Indigenous Peoples and local communities. The Ocean Biodiversity Information System (OBIS) and the Global Ocean Observing System (GOOS) are increasing the integration of local and Indigenous knowledge systems into ocean observation. The Biodiversity and Ecosystem Services Network (Bes-Net) explicitly values Indigenous and local knowledge on par with western science in its assessment processes. The Pacific Ocean Portal provides information on physical oceanography and traditional voyaging/navigational safety. The Pacific Climate Change Portal documents traditional knowledge relating to cyclones and sea-level rise. The Pacific Islands Protected Areas

Portal (PIPAP) maps locally managed marine areas, recognizing traditional tenure over government-designated marine protected areas. The Association of Southeast Asian Nations (ASEAN) Heritage Parks clearing-house provides data, frameworks and shared experiences to manage the protected areas collaboratively among the local communities and other stakeholders. The Convention for the Protection, Management and Development of the Marine and Coastal Environment of the Eastern African Region (the Nairobi Convention) clearing-house considers community-based coastal management as a key pillar of the implementation of the Nairobi Convention.

V. Examples of practices on sensitive and confidential information, cyber security and data sovereignty

47. The BBNJ Agreement requires the Mechanism to be “primarily of an open-access platform” (article 51, paragraph 2), while also requiring that the confidentiality of information provided under the Agreement and rights thereto be respected (article 51, paragraph 6). In practice, this requires a balance between open access and appropriate safeguards for confidential information. Furthermore, cyber security considerations are relevant, including risks related to unauthorized access, breaches of confidential information, or and the manipulation of data and information. Finally, questions have also been raised on whether a fully federated system would shift certain responsibilities on data security to the Parties that would need to provide information to the Mechanism through their national information systems and/or any other stakeholder that would provide information.

48. Existing information systems and databases use a range of safeguards to manage sensitive or confidential information. For example, the International Maritime Organization (IMO) requires user registration to ensure that documents reflecting ongoing discussions are accessible only to authorized users. The Global Information Exchange System (GIES) supporting the Agreement on Port State Measures is only accessible to Parties, as it contains compliance and enforcement information. The clearing-house mechanisms under the Convention on Biological Diversity (CBD) and its Nagoya Protocol generally avoid storing confidential contractual information, but, instead, hosts internationally recognized certificates of compliance that link to national systems where the confidential information resides. Furthermore, those who submit information may also state “confidential” in free-text fields to identify sensitive information. Platforms such as the Global Biodiversity Information Facility (GBIF) and the Ocean Biodiversity Information System (OBIS) generally do not host confidential data. Meanwhile, they may use Darwin Core terms, such as “informationWithheld” and “dataGeneralization” to mask precise coordinates of endangered or sensitive species to ensure their safety. Data providers may also use licensing options, such as CC0 or CC-BY, to maintain control over the data that is publicly released.

49. Other databases, information systems and clearing house mechanisms also incorporate institutional safeguards to support data sovereignty. For the three clearing-house mechanisms under the CBD and its protocols, information generally becomes an official national record only after validation by the relevant national focal point. Similar approach is also used in information systems, such as the focal point network established under the United Nations Convention against Cybercrime, which creates a secure channel accessible only to designated governmental representatives for the sharing of evidence.

VI. Multilingualism

50. Regarding multilingualism, the managers of various information systems, databases, and clearing-house mechanisms noted the need for caution on the use of artificial intelligence for the translation of technical and legal documents. Although the capabilities of artificial intelligence are improving rapidly, concerns remain regarding

accuracy and precision in highly technical or legally sensitive contexts. At the same time, a fully human-centered approach to translation, depending on the volume of documentation, may be resource-intensive. For example, the International Maritime Organization (IMO) operates in three working languages (English, French and Spanish) and six official languages. The organization generates two to three thousand documents annually and employs between thirty to fifty translators to support this workload.

51. The Global Biodiversity Information Facility (GBIF) currently supports twelve languages through its portal: Arabic, Chinese (Simplified and Traditional), Czech, English, French, Japanese, Polish, Portuguese, Russian, Spanish and Ukrainian. Translations are primarily produced by a global network of volunteer translators. GBIF also provides a service for national and thematic nodes to build their own data portals, which can be customized to support local languages. Technical documentation and training materials are frequently translated into languages such as French and Spanish to support capacity-building in different regions.

52. The Ocean Biodiversity Information System (OBIS) adopts a hybrid approach. Its primary information system and interface are predominantly available in English. However, similar to GBIF, OBIS operates through a network of national and regional nodes, many of which provide portals and user support in local languages to serve their local scientific communities. Some documents, including training materials, are often available in the official UN languages.

VII. Cost-benefit analyses

53. Based on desk-top research and interviews, the costs associated with the design, build, maintenance and improvements of information systems are generally difficult to determine precisely, given that they typically evolve over time. For example, the Global Plastic Hub, a platform recently developed by the United Nations Environment Programme (UNEP) required an initial investment of approximately USD 2,000,000 to develop core functionalities and contents, such as compiling knowledge materials and preparing data layers. Annual maintenance costs are estimated at approximately USD 120,000, covering personnel and operational expenses, including partial time contributions from a staff member at P-3 level (25 per cent); a national United Nations Volunteer (100 per cent), and a software engineer consultant (30 per cent), as well as hosting cost.

54. The Global Biodiversity Information Facility (GBIF) provides an example of a more complex platform for which information on operational costs is more readily available. However, because the platform was developed iteratively over the last 18 years, precise figures for the total costs of its design, build and maintenance are not available. It has been estimated that, for 2023, the core budget of the organization was approximately EUR 4 million, of which an estimated EUR 1 million to EUR 1.5 million supported the development and operation of its digital infrastructure, depending on how activities are categorized. While the GBIF Work Programme for 2023 does not publicly provide budgetary documentation, staff time allocations indicate that the priority areas include infrastructure and data (31 per cent), community and capacity (29 per cent), science and research (27 per cent), and policy and partnership (13 per cent).² This distribution suggests that expenses associated with infrastructure and data represents only part of its overall expenditure. In addition to the core budget, GBIF receives supplementary funding from external partners, including the Biodiversity Information

² The report is available at <https://docs.gbif.org/2023-work-programme/en/gbif-work-programme-2023.en.pdf>.

for Development programme of the European Union, which supports specific project-based activities.³

55. The European Bioinformatics Institute of the European Molecular Biology Laboratory (EMBL-EBI) provides another example of a large-scale database and an information system. It maintains a broad range of freely available biological data resources that enable scientists to study genes, proteins and molecular structures, and it also conducts computational research and provides specialized training in bioinformatics. EMBL-EBI is one of the three founding members of the International Nucleotide Sequence Database Collaboration (INSDC). According to an independent evaluation, the total annual investment required to develop and maintain the forty-four open data resources managed by EMBL-EBI is estimated at approximately GBP 110 million per year. Many of its data resources are developed through a combination of institutional funding and collaborative arrangements, including joint grants, consortium agreements (e.g. Protein Data Bank in Europe), direct partnerships (e.g. Reactome), and federated models (e.g. European Genome-phenome Archive). Some data resources are jointly managed with partner institutions (e.g. UniProt). Of the estimated GBP 110 million annual investment, approximately GBP 78 million represents direct operational expenses, while the remainder reflects financial and in-kind contributions from external collaborators and partners.⁴

56. GBIF and EMBL-EBI have also both undergone independent cost-benefit assessments.⁵ The results show that both platforms generate significant returns on investment, though they differ in scale, specific economic methodologies, and the magnitude of financial figures reported. The assessments also identified several common findings. First, both platforms were assessed as providing value relative to their cost. Second, time savings of researchers were identified as a primary driver of economic value. Third, both assessments adopted a counterfactual approach, asking users whether their work could be conducted in the absence of the platform. Finally, the assessments highlighted broader benefits, including support for the Sustainable Development Goals in the case of GBIF and strategic research and development impacts, including rapid responses to global challenges such as the COVID-19 pandemic, in the case of EMBL-EBI. A comparison of selected indicators from the two assessments is provided in Table 4 below.

Table 4.

Comparison of selected indicators from cost–benefit assessments of EMBL-EBI and GBIF

<i>Selected Indicators</i>	<i>EMBL-EBI Assessment (2021)</i>	<i>GBIF Assessment (2023)</i>
Annual operational cost	GBP 110 million	EUR 15 million
Direct user value	GBP 1.25 billion (estimated willingness to pay)	EUR 48 million (direct benefits)
Efficiency or time savings	GBP 2.6 billion – GBP 11 billion	EUR 35 million (approximately 845,000 hours)
Cost to benefit ratio	1 : 11 (minimum) to 1 : 102 (maximum)	1 : 3 (user) to 1 : 12 (societal)
Critical dependency	58 per cent could not obtain data elsewhere	47 per cent could not perform their work without it
Unique indicator measured	Avoided duplication of effort (GBP 6 billion value)	Volunteer contributions (EUR 958,000 value)
Societal or research and development impact	GBP 2.2 billion annual research and development return	EUR 185 million indirect societal benefits

³ Information about the funding is available at <https://www.gbif.org/news/4qE7yWlQqUkyGMYVD7Yr4C/eu-renews-funding-for-biodiversity-information-for-development-programme>.

⁴ The report by Charles Beagrie Ltd. is available at <https://www.embl.org/documents/wp-content/uploads/2021/10/EMBL-EBI-impact-report-2021.pdf>.

⁵ The report of GBIF can be obtained at: <https://www.deloitte.com/au/en/services/economics/perspectives/total-economic-value-open-access-database-living-world.html>.

57. The examples above illustrate that large-scale data-sharing platforms can generate substantial benefits relative to their operational costs. While the platforms examined may not be directly comparable to the Mechanism, they nevertheless demonstrate the potential value of investing in information-sharing platforms that build upon existing platforms and networks. The BBNJ Agreement also requires that the Mechanism build on global, regional and subregional clearing-house institutions, where applicable, when establishing regional and subregional mechanisms under the global mechanism (article 51, paragraph 3, subparagraph (d)). It will be essential, however, to prioritize which systems should be integrated at the initial stage and which could be incorporated incrementally. Furthermore, the Mechanism may benefit from developing partnerships with existing platforms and exploring opportunities for external support, such as joint projects and in-kind contributions.

VIII. Considerations for part (b)

58. The above stock-take study indicates that a critical consideration in the design and management of the Mechanism is not only the availability of data and information, but also the ability of Parties and other stakeholders to access, interpret and apply such information in a coherent and decision-relevant manner. The review of existing systems highlights the importance of building upon existing platforms and practices where appropriate. Informed by the stock-take study, the compilation of options to address technical requirements for the Mechanism in Part (b) focuses on not just what, but how, the information is provided in a tiered approach at Appendix A. Further, lessons learned from the existing databases, information systems and clearing-house mechanisms discussed in part (a) is reflected in the compilation of options in part (b).

Part (b)

A compilation of options to address technical requirements for the Mechanism

I. Introduction

59. Part (b) of the Study presents three indicative options for the operationalization of the Mechanism. Each option is described as an end-to-end model, covering the architecture, operating model, and oversight and management arrangements, and is assessed against a common set of requirements, such as accessibility for developing States Parties, including small island developing States, considerations regarding safeguards for traditional knowledge of Indigenous Peoples and local communities, the implementation of the “BBNJ” standardized batch identifier, and safeguards for confidential information. The options differ in the extent and timeframe in which each baseline requirements defined below in Section II may be addressed, as well as in their potential risks and challenges. Where relevant, the Study draws on findings from the stock-take study in Part (a). The objective of Part (b) of the Study is to facilitate the comparison of viable models on a consistent basis and to support consideration of a pathway that is feasible and scalable over time.

II. Overview of baseline requirements common to all options

60. The Options A through C and possible hybrid configurations presented in section V of Part (b) below assume a common baseline of required functionalities and quality or safeguard requirements, such as accessibility, security, confidentiality and interoperability, which are collectively referred to as “baseline requirements”. The requirements discussed in Sections II and III of Part (b) constitute the baseline requirements applicable to all options, unless otherwise indicated.

61. The following basic functionalities are required for each option for the operationalization of the Mechanism, while functionalities specific to the substantive elements addressed by the Agreement, which are discussed in paragraph 62 below, would operate on top of these basic functionalities:

- (a) Submission and publishing, including structured submission forms, document upload and version control;
- (b) Search and retrieval, including basic indexing, filtering, stable record pages and persistent links;
- (c) Notifications and alerts, including subscription-based alerts, deadline alerts and summary digests;
- (d) User management capabilities, including role-based access control, audit logging and account lifecycle management;
- (e) Confidentiality controls, including classifications such as public, restricted and confidential, together with appropriate enforcement mechanisms; and
- (f) Reporting and data export, including basic export functionalities for Parties and subsidiary bodies (e.g., CSV, JSON, PDF, Word and Excel formats).

62. Functionalities specific to the substantive elements addressed by the Agreement include the following:

- (a) Functionalities specific to marine genetic resources, including the fair and equitable sharing of benefits, including:
 - (i) Receipt, storage and searchability of pre-collection notification, and the triggering of automatic alerts when a pre-collection notification is submitted or updated;
 - (ii) Basic procedural checks of pre-collection notifications (e.g., automated completeness or schema checks, in accordance with modalities as may be determined by the Conference of the Parties) and the generation of the “BBNJ” standardized batch identifier upon receipt of a pre-collection notification that meets those requirements;
 - (iii) Receipt, storage and searchability of notifications of material changes to pre-collection notifications;
 - (iv) Receipt, storage and searchability of post-collection notifications and notifications at the stage of utilization;
 - (v) Receipt, storage and searchability of information received from Parties, including measures taken and national focal points;
 - (vi) Receipt, storage and searchability of periodic reports submitted by the Parties to the Access and Benefit-Sharing Committee regarding activities with respect to marine genetic resources and digital sequence information in accordance with modalities as may be determined by the Conference of the Parties;
 - (vii) Receipt, storage and searchability of information on traditional knowledge associated with marine genetic resources in areas beyond national jurisdiction in accordance with modalities as may be determined by the Conference of the Parties; and
 - (viii) Functionalities that support collaboration, such as structured submissions, versioned document uploads, threaded comments, automated alerts or deadline reminders.
- (b) Functionalities specific to area-based management tools, including marine protected areas, including:

- (i) Receipt, publication and searchability of public notifications relating to area-based management tools, including marine protected areas, and associated documentation, in accordance with modalities as may be determined by the Conference of the Parties, such as proposals and supporting documentation, consultation materials, adopted measures and relevant updates, including basic geographic tagging, where applicable, and alerts for key milestones and deadlines; and
- (ii) Functionality that supports collaboration, which may include role-based access for Parties and relevant instruments, frameworks and bodies, as appropriate, versioned document uploads, threaded comments or question-and-answer features during consultations, and automatic milestone or deadline alerts.
- (c) Functionalities specific to environmental impact assessments, including:
 - (i) Receipt, publication and searchability of information on screening, including decisions that an environmental impact assessment is not required for planned activities;
 - (ii) Receipt, publication and searchability of information on views registered by Parties on potential impacts of a planned activity and recommendations of the Scientific and Technical Body;
 - (iii) Receipt, publication and searchability of notifications of planned activities ;
 - (iv) Receipt, publication and searchability of opportunities for participation in the environmental impact assessment processes;
 - (v) Receipt, publication and searchability of draft environmental impact assessment reports made available during the public consultation processes;
 - (vi) Receipt, publication and searchability of information for stakeholders in the monitoring, reporting and review processes;
 - (vii) Receipt, publication and searchability of notifications issued regarding concerns registered by Parties that the authorized activity may have significant adverse impacts and any recommendations by the Scientific and Technical Body;
 - (viii) Receipt, publication and searchability of environmental impact assessment reports;
 - (ix) Receipt, publication and searchability of decision documents on planned activities;
 - (x) Receipt, publication and searchability of monitoring reports;
 - (xi) Receipt, publication and searchability of notifications of significant adverse impacts of authorized activities;
 - (xii) Receipt, publication and searchability of reports on the review of the impacts of authorized activities and decision documents on changes to authorization decisions; and
 - (xiii) Functionalities that support collaboration, such as structured submissions, versioned document uploads, threaded comments, automated alerts or deadline reminders.
- (d) Functionalities specific to capacity-building and the transfer of marine technology, including:
 - (i) Facilitation of the assessments of needs and priorities of developing States Parties and the matching of capacity-building needs with the support available and with providers of the transfer of marine technology, which may include enabling Parties and stakeholders to articulate capacity-building needs through supporting

templates, guidance and training, make public needs and offers, and make them discoverable through filters, keyword search and summary digests, including opportunities for training and technical assistance and relevant points of contact, with subscription-based alerts to reduce user burden; and

(ii) Functionality that support collaboration, such as structured submissions, versioned document uploads, threaded comments, automated alerts or deadline reminders.

(e) Functionalities specific to cross-cutting issues, including:

(i) Provision of core secretariat information and user support services, including notices of meetings, calendars and timelines, official documentation and templates relevant to Agreement-related workflows;

(ii) Publication, maintenance and searchability of documentation of the Conference of the Parties;

(iii) Publication, maintenance and searchability of information on documentation of the subsidiary bodies;

(iv) Functionality that support collaboration, such as structured submissions, versioned document uploads, threaded comments, automated alerts or deadline reminders; and

(v) Publication, maintenance and searchability of information on financial support, such as trust funds, funding opportunities, calls for support, eligibility and application information, and deadlines, including alerts for time-bound opportunities.

63. The following aspects are also included in the baseline technical requirements of each option and are discussed in section III of Part (b) below:

(a) Technical design requirements, such as ensuring that the Mechanism can work in areas with low connectivity or other access constraints, targeted alert systems, including proximity-based notification systems, and other characteristics as requested by delegations at the first and second sessions of the Preparatory Commission;

(b) The generation and use of the “BBNJ” standardized batch identifier to operationalize the relevant provisions of Part II of the Agreement;

(c) The most appropriate user system with differentiated access, possible verification and validation processes to achieve transparency in view of the objective of the Agreement;

(d) The role that regional or subregional nodes could play in the Mechanism;

(e) Ensuring accessibility, including through a user-friendly interface, including tools for uploading and downloading notifications required under the Agreement;

(f) Safeguards for the management of confidential information;

(g) Ensuring that the sharing of traditional knowledge of Indigenous Peoples and local communities is done appropriately, including recognizing and respecting free, prior and informed consent and other relevant rights associated with that knowledge, particularly those of Indigenous Peoples, including in light of lessons learned on how other databases, repositories, etc., have addressed sharing of traditional knowledge and protection; and

(h) Enabling interoperability with clearing-house mechanisms and databases of relevant instruments, frameworks and bodies, including at the subregional level.

64. The consideration of any other technical design requirements that may arise in the operationalization of the functions of the Mechanism, including in relation to specific

modalities for the operation of the Mechanism that may need to be determined by the Conference of the Parties at its first meeting is addressed in Section VI of Part (b).

III. Baseline requirements for the aspects outlined in the terms of reference for the Study

Targeted alert systems, including any proximity-based notification systems

65. The Mechanism could incorporate three alert modes as built-in features, enabling users to select the option that best suits their needs in a given use case. Implementing and maintaining these modes may entail different levels of cost and technical complexity:

(a) Subscriptions by topic: users subscribe to substantive elements, themes, keywords, tags and procedural stages, such as the scoping stage during an environmental impact assessment, the publication of a draft environmental impact assessment report or monitoring reports;

(b) Subscriptions by geography (proximity-based): alerts are triggered when an activity footprint intersects a user-defined area of interest or buffer zone, such as adjacent zones or other configured polygons; and

(c) Digests: weekly (or configurable) summary reports tailored to user subscriptions, aimed at reducing human workload and information overload.

66. Minimum viable proximity-based alerts could initially be implemented through simple geofencing (areas of interest defined as polygons with buffered intersections) and progressively enhanced through improved geospatial tooling, higher-resolution boundaries, and performance optimization.

Accessibility requirements (including low connectivity)

67. The Mechanism could be designed to support use under low-bandwidth conditions, including minimal user interactions, lightweight pages, and text-first navigation options. To avoid centralizing access and to ensure usable performance for geographically remote users, including those in small island developing States, the Mechanism could be hosted on off-premise, cloud-based infrastructure with multi-region availability and a content-delivery layer (e.g., regional mirror servers or edge nodes, similar to those used by content delivery networks). This would allow public content and core services to be delivered from locations closer to users, thereby reducing latency and improving resilience in the event of disruptions affecting any single location. This hosting posture could include high availability and disaster-recovery capabilities as baseline design assumptions.

68. The Mechanism could also support offline continuity for data entry, at a minimum through “save as draft” functionality and template-based offline completion, and, where feasible, through queued submissions with synchronization when connected. Workflows could be mobile-tolerant so that forms and uploads are usable on commonly available devices. Multilingual navigation and core workflows could be provided, together with a clear translation policy for submitted documents. The Mechanism could also provide user support through a helpdesk function (e.g., ticketing and email support with scheduled support hours), complemented by training materials and guided walkthroughs. These features would be in line with the practices discussed in Part (a) of the Study.

“BBNJ” standardized batch identifier

69. The Mechanism will generate a “BBNJ” standardized batch identifier upon receipt of a pre-collection notification. This could be done together with a limited, standardized set of core metadata (such as the submitting Party or entity, the date of issuance and a high-level description of the planned collection) sufficient to allow the identifier to be resolved, understood and linked to related records within the Mechanism.

70. The “BBNJ” standardized batch identifier could be envisaged as a globally unique, persistent, resolvable and machine-readable identifier, which aligns with the FAIR Principles (i.e., findable, accessible, interoperable and reusable). It could be designed as a stable identifier whose structure does not change over time, so that it can be referenced consistently in external scientific databases and repositories (for example, as a citation or metadata field) without requiring those systems to modify their own identifier schemes or workflows. Within the Mechanism, the identifier would link the pre-collection notification with the post-collection notification and any subsequent notifications at the stage of utilization. The Mechanism could support version control, allowing changes to the underlying metadata (e.g., updates to a cruise plan) without compromising the integrity or persistence of the identifier.

71. Different technical approaches to the generation and management of the “BBNJ” standardized batch identifier within the Mechanism are possible, and an illustrative example is provided for options A and B further discussed below. For option C, the technical approaches to the generation and management of the “BBNJ” standardized batch identifier may depend on the existing system to be retrofitted.

User system, verification and validation (transparency by design)

72. The Mechanism could support differentiated user roles and implement verification and validation measures to ensure transparency and reliability. A baseline set of roles could include, for example, secretariat administrators, national focal points, designated Party users, authorized traditional knowledge account holders, authorized observers, and public users (with read-only access to public content). The differentiation of user roles would be subject to decisions by the Conference of the Parties.

73. Verification of user accounts could include the designation of official users at the national level whereby Parties would nominate and approve their own authorized accounts, optional domain-based checks such as institutional email verification, and two-factor authentication for users with higher level of system privileges.

74. Validation procedures for submitted information could include:

- (a) Automated checks, such as verification of required fields, format checks, geometry checks for spatial data, and malware scanning, complemented by human review for completeness and appropriate classification;
- (b) Optional national approval steps, in particular where domestic rules require Party approval before information is published; and
- (c) Clear indication of the status of each submission, for example, whether it is in draft form, submitted, published or restricted.

75. To further support transparency, the Mechanism could include additional tools, such as time stamps, version histories, change logs for non-confidential records, and audit logs for access to restricted content.

Role that regional or subregional nodes could play in the Mechanism

76. Subject to decisions by the Conference of the Parties, regional or subregional nodes could play a role in supporting the operation of the Mechanism, particularly in relation to accessibility, user support, outreach and facilitation in low-connectivity or capacity-constrained contexts. Such nodes could, for example, assist users in preparing submissions, provide training and guidance, or support awareness-raising at regional or subregional level.

77. Regional or subregional nodes could also support the provision of links to relevant regional and subregional clearing-house mechanisms and other gene banks, repositories and databases, in accordance with Article 51, paragraph 3, subparagraph (c) of the BBNJ Agreement. In addition, such nodes may be relevant to the implementation of Article 51,

paragraph 3, subparagraph (d) of the BBNJ Agreement, which provides that the Mechanism shall build on global, regional and subregional clearing-house institutions, where applicable, when establishing regional and subregional mechanisms under the global mechanism.

78. Any use of regional or subregional nodes would be subject to agreed oversight and management arrangements and would not necessarily imply decentralization of authority for the validation, publication or maintenance of the authoritative record, which could remain centralized.

Uploading and downloading toolkit for notifications and records

79. To ensure usability, the Mechanism could include tools for both uploading and downloading notification-related information. Possible features could include:

(a) Upload: structured submission forms for different notification types; options for bulk upload where repeated submissions are required; and attachment management, including file size controls and compression options;

(b) Download (human-readable formats): the ability to download individual records in formats such as PDF, Excel or Word, to download attachments, or to download complete case-file bundle as compression files;

(c) Download (machine-readable formats): the ability to export record in formats such as JSON; bulk export by date range or package element; subscription feeds (e.g., RSS, Atom or email digests); and, where applicable, authenticated application programming interfaces for authorized system-to-system information exchange; and

(d) Integrity safeguards: mechanisms such as checksums for downloaded bundles and stable version histories for each record.

Safeguards for the management of confidential information

80. All options could incorporate a consistent confidentiality framework that balances the primarily open-access nature of the Mechanism with the need to protect confidential or proprietary information and rights related thereto. Possible elements of such a framework could include:

(a) Classification of information at the time of submission, using public, restricted, and confidential or proprietary classifications, supported by a rationale field and, where appropriate, an expiry or review date;

(b) Enforcement of access controls through role-based permissions with least-privilege defaults, including the separation of confidential storage from public information delivery paths;

(c) Security measures such as encryption in transit and at rest, secure file handling, malware scanning, and periodic access reviews;

(d) Redaction workflows allowing the publication of public versions of records while maintaining confidential or proprietary originals in a restricted environment;

(e) Audit mechanisms, including immutable access logs of access to restricted or confidential information, as well as incident response and breach notification procedures;

(f) Retention policies defining archiving policies for confidential or proprietary information and secure deletion procedures where required; and

(g) Protection of personal data by applying privacy-by-design and data-minimization approaches, together with alignment with recognized international information security standards, such as ISO/IEC 27001 and ISO/IEC 27701 or equivalent, consistent with any policies that may be applicable to the secretariat of the

Agreement, which will manage the Mechanism, depending on the arrangements for its functioning.

Safeguards for traditional knowledge, including free, prior and informed consent

81. All options would ensure that traditional knowledge of Indigenous Peoples and local communities is handled appropriately. Drawing on the practices discussed in Part (a) of the Study, the Mechanism could incorporate the following safeguards:

- (a) Structured recording of consent and rights-related information, including free, prior and informed consent, approval or involvement, and any conditions governing the access to such knowledge;
- (b) Application of data-minimization approaches, including the option to create metadata-only records with controlled access pathways but without requiring public disclosure of the underlying traditional knowledge;
- (c) Alignment of oversight and management arrangements with Indigenous data governance principles, such as the CARE Principles for Indigenous Data Governance and support community-appropriate processes;
- (d) Implementation of access controls tailored to traditional knowledge, including permission management, audit logs and controlled access request workflows; and
- (e) Adoption of a “do-no-harm” design approach that avoids default public disclosure of traditional knowledge and ensure that access is governed by consent and agreed conditions.

Interoperability with other databases, information systems and clearing-house mechanisms

82. Interoperability with other databases, information systems and clearing-house mechanisms could be supported through a tiered approach. All interoperability arrangements with external mechanisms and databases would need to be implemented in accordance with applicable standards for information and communication technologies, such as those of the United Nations, including in relation to software, hardware and information security. Possible interoperability modes could include:

- (a) Link mode: provision of authoritative links to external systems, supported by consistent metadata and stable identifiers;
- (b) Harvest mode: periodic harvesting of metadata, where permitted by partners, allowing the Mechanism to adapt to existing partner protocols without requiring changes to those systems;
- (c) Application programming interface federation: deeper system-to-system integration through authenticated interfaces, subject to agreed oversight and management arrangements and security controls.

83. Taken together, these modes could improve discoverability, consistency and efficiency across systems and, where appropriate, help reduce duplicative reporting by enabling data and information submitted in one system to be referenced or reused by others.

84. Interoperability could be supported at global, regional, subregional and sectoral levels through agreements such as memoranda of understanding, and through the use of common standards, including shared metadata profiles, vocabularies and identifiers.

External information sources integration framework

85. Two categories of information are relevant to the Agreement. First, the implementation of the Agreement will generate information and documents, and the Mechanism will function as a centralized platform to enable Parties to access, provide

and disseminate such information. Second, the Agreement requires that the Mechanism provide links to relevant global, regional, subregional, national and sectoral mechanisms and other gene banks, repositories and databases (including those pertaining to relevant traditional knowledge of Indigenous Peoples and local communities), and promote, where possible, links with publicly available private and non-governmental platforms. In practice, these requirements could lead to a hybrid approach combining centralized Agreement-related workflows with a federated approach that builds on existing databases, information systems, and clearing-house mechanisms.

86. As discussed in Part (a) of the Study, a critical design consideration will be, first, to determine which external information sources should be integrated at the initial stage of operationalization and which should be integrated at later stages; and second, to define, for each prioritized source, the appropriate access modality and interoperability approach. These choices have implications for usability, oversight and management, cost and lead time, as further discussed below.

(a) Access modality (i.e., where the data and information reside): To support clear communication with users who may not be familiar with data system design, the Mechanism could clearly describe how external data and information is made available through it. In this regard, a key distinction is between data and information generated pursuant to the Agreement and data and information that already exists elsewhere. Depending on where the data and information reside, the Mechanism could apply the following access modalities:

(i) Hosted in the Mechanism. The Mechanism stores and provides access to data and information generated pursuant to the Agreement, which may include, records, documents and, where appropriate, curated information products. This modality provides highest level of control and continuity, with higher operational responsibility, including storage, curation, version control, security and long-term maintenance.

(ii) Accessed through federated queries or application programming interfaces: The Mechanism retrieves data and information from external databases, information systems and clearing-house mechanisms through defined interfaces, while the authoritative data remain in the external systems. This modality may improve user experience and reduce duplication of data, but may require formal agreements, technical standards, secure access arrangements, and ongoing coordination.

(iii) Referenced through metadata and external links: The Mechanism provides a stable record page containing key descriptive metadata and an authoritative link directing users to the external source where the information is stored. This modality has the lowest technical dependency and is fastest to implement, but it relies on the continued availability of external databases, information systems and clearing-house mechanisms and may provide more limited integrated search and analytics, unless supplemented by indexing.

(iv) Represented through derived products or indicators: The Mechanism provides derived summaries, such as contextual layers, indicators and screening products, produced from one or more sources, rather than replicating the underlying datasets. This modality could support decision-making and reduce user burden, but requires transparent provenance, reproducibility expectations and careful treatment of licensing and sensitivities.

(b) Interoperability Modes (i.e., how systems connect): The Mechanism could apply a progressive approach to interoperability. These modes describe the technical pattern used to connect systems and may be applied across the access modalities described above. For example, referenced sources are typically implemented through link mode, while federated sources rely on application programming interfaces. The

modes listed below are presented in the order of least technically demanding to more advanced.

- (i) Link mode (baseline): Authoritative links to external systems are provided through consistent metadata, stable identifiers and clear provenance on Mechanism record pages. This mode can enable initial connectivity with minimal technical complexity and low burden for partner systems.
- (ii) Harvest mode (selective metadata harvesting): The Mechanism periodically collects metadata from external systems to support unified search and discovery, while authoritative records remain in the source systems. This mode allows users to search across multiple connected systems from a single interface, reducing the need to navigate separate platforms.
- (iii) Application programming interface federation (deeper system-to-system integration): Authenticated and governed integrations through application programming interfaces allow deeper technical connections, which can support higher-value use cases, such as identifier resolution, automated cross-referencing, or controlled exchanges for authorized users, subject to registration, change control, and security requirements. This mode is typically used for targeted integrations where benefits justify the ongoing effort to operate, maintain, and govern the connection over time.

87. The Mechanism could integrate external information sources progressively, using a transparent prioritization approach that supports a scalable “system-of-systems” model. Phasing reflects both priorities and the relative complexity of integrating different categories of external sources, including oversight and management requirements, access constraints and implementation effort. While the same tiered structure applies across all three options, the speed and scale of implementation may differ. Complexity increases from Tier 1 to Tier 3:

(a) Tier 1: The objective of Tier 1 would be to establish immediate, low-risk connectivity to widely used external data and information sources that support key functions of the Mechanism or decision-making under the Agreement, such as those supporting notifications on activities with respect to marine genetic resource and digital sequence information, contextualizing the establishment of area-based management tools and screening during environmental impact assessments, and facilitating capacity-building needs and priorities assessment.

- (i) Default interoperability approach: link mode for Tier 1 sources, with limited and selective metadata harvesting pilots where appropriate and explicitly permitted by partners.
- (ii) Typical characteristics: open access or low operational friction; use of stable and well-established identifiers and metadata practices in the source systems; and high levels of scientific or community usage.

(b) Tier 2: The objective of Tier 2 would focus on integrating external databases and information systems where access is conditional or require formal arrangements, or where technical integration depends on externally governed application programming interfaces or data schemas.

- (i) Default interoperability approach: link mode as an initial step, followed by metadata harvesting or application programming interface federation once required agreements, technical protocols and oversight and management arrangements are in place.
- (ii) Typical characteristics: access controlled by the data owner; restrictions on copying or automated retrieval of data; confidentiality or mandate limits apply; and integration requiring ongoing coordination within system owners.

(c) Tier 3: The objective of Tier 3 would involve the integration of quasi-commercial or licensed information sources and other more complex services where the potential benefits are substantial, but integration requires licensing, partnerships, and/or sustained operational investment.

(i) Default interoperability approach: link mode initially, with progression to licensed application programming interface federation or the provision of derived products or indicators, where permitted by contractual and policy arrangements.

(ii) Typical characteristics: subscription or licensing terms; variable access conditions; potentially high operational and procurement costs; explicit decisions by the Conference of the Parties on oversight and management arrangements, as well as on resourcing, required.

88. To avoid repeating database inventories across the option narratives, the Mechanism could maintain a single non-exhaustive external source integration matrix. The matrix could set out:

(a) Recommended external sources for each tier;

(b) The intended access modality, as discussed in paragraph 86(a) above;

(c) The intended interoperability mode, as discussed in paragraph 86(b) above; and

(d) Key prerequisites, such as standards and identifiers, required agreements, licensing arrangements, security controls, update cadence, and partner change control arrangements.

89. An indicative and non-exhaustive external source integration matrix is available in Appendix A.

IV. Indicative options for the operationalization of the Mechanism

90. The options presented in this Study, and the hybrid configurations, are intended to support decisions on how the Mechanism may be overseen, managed and scaled over time. They describe different oversight and management arrangements, including on how responsibilities are allocated, how participation is supported, and where authoritative control sits.

91. The three options are presented as complete profiles. Each option combines two sets of choices: first, oversight and management arrangements (i.e., a central system only, or a central system complemented by nodes); and second, delivery modalities (i.e., a system built specifically for the Mechanism, or one based on retrofitting an existing platform). These choices are interlinked and have practical implications for responsibilities, capacity requirements, risk management, phasing and delivery arrangements. The options are designed to facilitate comparison of how the Mechanism could be developed and function in practice, while allowing for hybrid configurations.

92. The options do not prescribe a fixed IT architecture or detailed system design. Rather, they define the operating constraints and service expectations that guide engineering decisions, such as how data submission and validation workflows are implemented, how notifications are processed and delivered, how access controls are applied, how systems are hosted and secured, and how integrations are managed over time. For example, more centralized arrangements may be appropriate where uniform safeguards, consistent application of rules and clear accountability are required. More distributed arrangements may be appropriate where participation barriers are significant and closer support to users improves effectiveness, provided that the official record remains coherent. Retrofitting existing platforms may be appropriate where required functionalities can be adopted without constraining safeguards or workflows.

93. From an IT architectural perspective, the Mechanism will primarily be implemented through a set of shared platform capabilities that operate across the baseline requirements, but are configured to meet function-specific requirements and safeguards. These capabilities include user roles and access control; structured submission, validation and publication workflows; search and retrieval; notifications and alerts; confidentiality handling and auditability; interoperability with external systems; reporting and export; and user support and operational services.

94. Hybrid configurations are included to reflect that the three options need not be applied across all functions of the Mechanism. Different capabilities may warrant different approaches, provided that responsibilities and lines of authority remain clear and the Mechanism functions as a single authoritative system to meet the baseline requirements. The illustrative example of a hybrid configuration is therefore a structured way to express preferences about responsibilities and lines of authority and support, rather than a menu of independent IT architectural features.

95. Selecting option A, B or C, or a hybrid configuration, does not imply prioritization of one substantive element function over another, nor does it predetermine engineering decisions. Statements such as “prioritizing accessibility”, when translated into IT architectural design, would mean, for example, low-bandwidth design assumptions, offline continuity measures, user support arrangements, or alert delivery methods, and not when framed as a comparison between oversight and management arrangements. The baseline requirements are applicable to all options.

96. To summarize, the purpose of comparing options is to bridge decisions on oversight and management arrangements and delivery modalities, and to facilitate IT architectural planning. It provides the basis for defining scope, safeguards, service expectations, and operating constraints that a build team or vendor would work within, while leaving detailed system design to be developed and validated during the operationalization phases (i.e., Phases 1 through 4, see Part (c) of the Study). The options framework does not present a choice among fixed architectures, but provides a structured way to articulate choices among different oversight and management arrangements and delivery modalities, without requiring the specification of IT architectural details at this stage.

97. This section provides an overall summary of three indicative options for the operationalization of the Mechanism, accompanied by a comparative reference table. As noted above, elements of these options can be combined to form new options. Cost assumptions for the options are discussed in Appendix B.

Baseline requirements

98. All options are designed to meet the same baseline requirements as outlined in Section III of Part (b) above.

Differences across options

99. The options differ on how the baseline requirements are addressed, including with respect to:

(a) Where complexity resides (e.g., a single integrated Mechanism, a Mechanism designed to have support from regional or subregional nodes from the outset, or a Mechanism built on a retrofitted platform with external dependencies);

(b) When resources are required to introduce additional complexity (e.g., lower up-front build complexity with deeper interoperability introduced later, potentially requiring refactoring, compared with higher up-front build complexity to make deeper interoperability safer and easier to expand earlier);

(c) Who manages the Mechanism (e.g., the secretariat alone, the secretariat in cooperation with entities that manage regional or subregional nodes, or the secretariat

in cooperation with entities that manage an existing platform in the case of a retrofit option); and

(d) How each model scales as the number of Parties and authorized users increases, as submission and record volumes grow, and as consultation activity expands, including whether scaling is handled primarily through the expansion of the capacity of the secretariat, distributed support arrangements (i.e., nodes), or operating within the potential constraints of a retrofitted platform.

A simpler way to understand the options

100. The choice among these three options concerns less individual features and more responses to three practical questions:

(a) Where does the work take place? (i.e., the secretariat, the secretariat plus nodes, or through the retrofit of an existing platform);

(b) How are access and participation supported in practice? (i.e., a central helpdesk, regional and subregional support, or platform-based support); and

(c) How much change management is deferred to later stages? (i.e., introducing nodes later, operating nodes from the outset, or managing platform upgrades and constraints).

Factors not suited for feature-by-feature comparison

101. Certain requirements are reflected across all options, but their meaning and practical implications differ depending on the underlying operating model. In these areas, a simple feature comparison may create the misleading impression that one option has something another option lacks. Examples include:

(a) Accessibility and low-connectivity requirements: all options meet accessibility and low-connectivity requirement; differences relate to whether support is provided only by the secretariat or supplemented through regional or subregional nodes;

(b) Confidentiality safeguards: all options apply varied access controls and audit mechanism; differences relate to where review, redaction and staging occur and how oversight is exercised across multiple operating environments;

(c) Safeguards for traditional knowledge: all options support consent-based, non-default publication; differences relate to whether oversight, management and facilitation can be supported closer to communities; and

(d) Interoperability: all options initially rely on the link mode discussed in paragraph 86(b) above; the options differ in how quickly more integrated connections, such as metadata harvesting or application programming interfaces can be introduced, which will also depend on how fast the required agreements, access controls and technical rules can be established.

Option A: Central system⁶ with minimal dependencies and progressive integration

102. Under this option, the Mechanism would be operationalized as a central system managed by the secretariat that delivers Agreement-related workflows with minimal external dependencies. Integration with external information sources would be deepened progressively over time, while most scientific and other information would continue to reside in existing external platforms.

103. The complexity of this option sits primarily within the secretariat, including core functions, user support and moderation functions. There would be lower complexity up

⁶ An example of an existing central system is the Global Integrated Shipping Information System (GISIS) of the International Maritime Organization.

front, with additional complexity introduced later if Parties subsequently decided to pursue deeper integration or introduce nodes.

104. Upon initial operationalization, this option would provide a single public entry point for the Mechanism delivering baseline capabilities, including:

- (a) Structured submissions through forms and document uploads, with versioning;
- (b) Search and retrieval functionalities, including basic indexing and filtering, and stable record pages and links;
- (c) Alerts, including subscriptions by topic or stage, deadline reminders and summary digests;
- (d) User roles and access control, audit logging and account management;
- (e) Confidentiality classification (public, restricted or confidential) with enforcement mechanisms;
- (f) Basic reporting and data exports for Parties and subsidiary bodies;
- (g) Generation of “BBNJ” standardized batch identifiers, including a resolver page with minimal metadata for lookup and linkage;
- (h) Low-bandwidth design features (e.g., lightweight pages and minimal user interactions) or basic offline continuity (e.g., draft saving and template-based offline completion for later upload); and
- (i) Centralized helpdesk support (e.g., ticketing and email support with scheduled support hours), as well as basic training and “how-to” materials.

105. Interoperability with external information systems would follow a tiered approach. Initial operationalization would rely on link mode, with selective expansion over time to introduce metadata harvesting or application programming interface-based integrations for a limited number of priority use cases, once appropriate oversight and management arrangements and security arrangements are in place.

106. Potential advantages of this option include:

- (a) Faster and simpler path to initial operational readiness;
- (b) Lower initial build complexity, with fewer technical dependencies;
- (c) Clear accountability through a single primary operator; and
- (d) Avoidance of duplicating large scientific datasets by building on existing system.

107. Potential trade-offs and risks include:

- (a) Higher early operational workload for the secretariat (e.g., moderation, publication quality assurance, and helpdesk demand during peak periods);
- (b) Some usability improvements being deferred to later phases;
- (c) The possibility that introducing nodes or deeper integrations later may require redesign of certain workflows rather than simple add-ons; and
- (d) Proximity-based alerts, if provided by the Mechanism, initially operating at a minimum viable level, with geospatial accuracy and performance requiring enhancement as usage grows.

108. Indicative resource requirements, expressed in full-time equivalent (FTE) workload estimates, for phases 1 through 4 (see Part (c) of the Study) are as follows:

- (a) Build and configuration (one-off): approximately 24 to 40 FTE-months (2 to 3.3 FTE-years);
- (b) Peak build and go-live period: approximately 4 to 7 FTE (concurrent staffing);
- (c) Ongoing operations (annual, secretariat-resourced): approximately 2.5 to 4.0 FTE per year; and
- (d) Distributed effort outside the secretariat (e.g., nominating users, reviewing and approving submissions, basic coordination across agencies, coordination with the secretariat, etc.): approximately 0.05 to 0.15 FTE per national focal point, plus delegated user time.

109. Decisions of the Conference of the Parties that would affect cost and scope under this option include minimum metadata and attachment requirements, multilingual policy, validation rules, confidentiality criteria, proximity-based alert policy, and the external databases and information systems to be integrated pursuant to the tiered approach.

Option B: Central system with regional and subregional nodes⁷

110. Under this option, the Mechanism would be operationalized through a secretariat-managed central system that remains the authoritative source for records generated pursuant to the Agreement, complemented by regional or subregional nodes that provide localized support, assisted workflows and improved connectivity.

111. This option distributes complexity across the central system and nodes, subject to common standards and oversight. It requires greater coordination and resources up front, intended to reduce later bottlenecks and improve resilience as number of Parties and authorized users and the corresponding number of submissions and records grow.

112. Upon initial operationalization, this option would deliver all baseline requirements described under Option A through the central system, with nodes that could provide:

- (a) Offline continuity through queued synchronization (i.e., preparing submissions offline and syncing when connected);
- (b) Local performance improvements, such as caching or mirroring public content closer to users;
- (c) Assisted validation workflows to reduce errors before records are submitted to the central system; and
- (d) Localized user support, including language and time-zone coverage, and regionally or subregionally appropriate support.

113. Under this option, the central system would remain the authoritative location for the publication of official records, the management of global alerts, and the completion of core Agreement-related workflows, including the generation of the “BBNJ” standardized batch identifiers.

114. Potential advantages of this option include:

- (a) Stronger practical support for developing States, in particular least developed countries and small island developing States, which may face connectivity and bandwidth constraints;
- (b) Improved quality and timeliness of submissions (e.g., fewer errors and more effective engagement) through localized support and assisted workflows;

⁷ Examples of such oversight and management arrangements include the Ocean Biodiversity Information System (OBIS) and the Global Biodiversity Information Facility (GBIF).

(c) Greater flexibility to support functions relating to traditional knowledge closer to communities (i.e., sensitive content can remain within trusted arrangements while metadata and access conditions are handled centrally); and

(d) Better long-term resilience and scalability if the Mechanism needs to accommodate higher-than-expected volume of use.

115. Potential trade-offs and risks include:

(a) Higher build and operating costs compared with Option A;

(b) Increased oversight and management requirements to ensure consistent standards and accountability across nodes;

(c) Risk of uneven service quality if node capacity varies; and

(d) Stronger coordination requirements for security and confidentiality controls across multiple operating environments.

116. Indicative resource requirements (FTE-based) for phases 1 through 4 (see Part (c) of the Study) are as follows:

(a) Build and configure (one-off): approximately 42 to 80 FTE-months (3.5 to 6.7 FTE-years);

(b) Peak build and go-live period: approximately 6 to 10 FTE;

(c) Ongoing operations (annual, secretariat-resourced core): approximately 4.5 to 7.5 FTE per year; and

(d) Distributed effort outside the secretariat: approximately 1 to 3 FTE per year per node (where deployed), plus national focal point effort (as under Option A).

117. Decisions of the Conference of the Parties that would affect cost and scope under this option include all decisions identified under Option A, as well as whether nodes are endorsed or optional, minimum standards applicable to nodes, who operates nodes, the extent to which restricted workflows may be handled by nodes locally (and under what controls), and expected support coverage and service levels.

Option C: Retrofit an existing clearing-house mechanism or an information system⁸

118. Under this option, the Mechanism would be operationalized by retrofitting an existing clearing-house platform to provide core capabilities (e.g., identity management, workflows, publication and versioning), supplemented by specific modules to perform functions required under the Agreement.

119. In this model, complexity is divided between the retrofitted platform, with its capabilities and constraints, and the management by the secretariat of its dependencies on that platform. Up-front efficiencies may arise if the platform aligns well with the requirements, while long-term costs and risks may arise from customization, upgrades and change-management requirements.

120. Upon initial operationalization, this option could deliver baseline capabilities more rapidly if the retrofitted platform already provides robust core functionalities, such as:

(a) User registration and role-based access;

(b) Submission workflows and templating;

⁸ An example of a retrofit approach is the Access and Benefit-Sharing Clearing-House (ABSCH) under the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity (CBD). The ABSCH was initially developed by building on the core functionalities of the Biosafety Clearing-House (BCH) under the Cartagena Protocol on Biosafety to the CBD, which had been built earlier.

- (c) Publication patterns and version history; and
- (d) Basic search and record pages.

121. Additional modules specific to the requirements under the BBNJ Agreement would still need to be developed or adapted to support:

- (a) The generation and resolution of “BBNJ” standardized batch identifiers;
- (b) Agreement-specific alerts, including proximity-based alerts if such a feature is provided;
- (c) Workflows relating to the establishment of area-based management tools, including marine protected areas;
- (d) Workflows relating to environmental impact assessments, including bundled document downloads;
- (e) Confidentiality controls and safeguards for traditional knowledge of Indigenous Peoples and local communities, including metadata-only publication and controlled access requests; and
- (f) Functionalities supporting the needs and priorities assessment of developing States Parties in terms of capacity-building and the transfer of marine technology, and the matching of capacity-building needs with the support available and with providers for the transfer of marine technology.

122. Potential advantages of this option include:

- (a) Reduced build time by avoiding the development of standard platform capabilities from scratch;
- (b) Ability to leverage existing maturity in security, identity management and workflow tooling; and
- (c) Potential simplification of training where the platform is already familiar to some users.

123. Potential trade-offs and risks include:

- (a) Cost savings are not guaranteed, as missing capabilities may require substantial customization;
- (b) Long-term sustainability depends on the roadmap and upgrade cycles of the retrofitted platform;
- (c) Risk of lock-in or reduced flexibility if the Mechanism becomes tightly coupled to the retrofitted platform; and
- (d) Ongoing operational workload remains, including helpdesk support and moderation.

124. Indicative resources requirements (FTE-based) for phases 1 through 4 (see Part (c) of the Study) is not available for this option.

125. Decisions of the Conference of the Parties that would affect the cost and scope under this option include all decisions identified under Option A, as well as selection of the platform to be retrofitted, the long-term operating model, requirements for geospatial and offline functionalities, the strictness of confidentiality and safeguards for traditional knowledge, and expectations regarding control and flexibility versus speed of deployment.

Table 5.
Indicative comparison of Options A to C

<i>Comparable decision point</i>	<i>Option A (Central system)</i>	<i>Option B (Central system + nodes)</i>	<i>Option C (Retrofit platform)</i>
Meeting baseline requirements upon initial operationalization	Yes	Yes	Yes (subject to the fitness of the selected platform)
Primary workload	Secretariat	Secretariat and nodes	Secretariat and/or existing clearing-house mechanism or information system
Initial operationalization complexity (relative)	Lower	Higher	Variable
Speed to initial operational readiness (relative)	Faster	Slower	Variable
Indicative build effort (FTE, one-year build)	~2 to 4	~4 to 7	N/A
Indicative annual operational effort (FTE per year, the secretariat core)	~2.5 to 4.0	~4.5 to 7.5	N/A
Additional effort outside the secretariat (structural)	national focal points (small fraction of FTE)	Nodes (1 to 3 FTE per node per year), plus national focal points (same as Option A)	N/A
Oversight and management burden (relative)	Lower	Higher (standards and assurance across nodes)	Medium–High (platform oversight and change-management)
Primary long-term risk	Central bottlenecks and potential later redesign if distribution is introduced	Coordination burden and uneven node capacity	Customization, upgrade dependency, or platform lock-in risk
Indicative suitability	Prioritizing rapid initial deployment and simplicity	Prioritizing early investment in distributed support and resilience	Prioritizing reuse of an existing platform, subject to sustained suitability

126. A table of elements that could be considered for hybrid configurations and an illustrative example of a hybrid configuration are presented in Appendix C below.

V. Artificial intelligence-assisted interaction layer (natural language and agentic interaction) as a top-layer to the selected option

127. An optional, additive capability in the form of an artificial intelligence (AI)-assisted interaction layer could be introduced, irrespective of which option is selected for the operationalization of the Mechanism. This layer would sit above the selected design of the Mechanism and is intended to reduce user burden and improve discovery across the content of the Mechanism and approved external information sources, without changing the underlying allocation of authority or the core workflows.

128. This layer could be implemented through controlled connectors (i.e., governed interfaces to the search and indexing functions of the Mechanism and to any approved external information sources). Such connectors could enable:

(a) Natural-language search and guided navigation, including multilingual query handling (e.g., allowing users to search in one language and retrieve relevant results stored in another), with clear references to the underlying records; and

(b) Authorized automated assistance (i.e., software tools operating solely on pre-defined instructions) to execute approved queries against the records of the Mechanism and against indexes or federated application programming interfaces, subject to access controls, audit logging and limits on request frequency.

129. The primary purpose of this layer would be to support discovery and navigation across large volumes of the content within the Mechanism and linked information sources, and to reduce manual effort for well-defined, repetitive information retrieval tasks, without altering decision-making authority. The layer would function as an overlay to the selected interoperability modes (link, metadata harvesting, or application

programming interface) and would only access information available under the selected mode. It would not bypass oversight and management arrangements or permissions. Outputs generated through this layer should be treated as user-support tools, rather than official or legally authoritative outputs under the Agreement. Any AI-enabled function would need to be auditable, respect confidentiality safeguards and safeguards for traditional knowledge, and follow a clear policy on where human verification is required, particularly for technical, legal or time-sensitive content where precision is essential.

130. Indicative additional resource requirements (high-level) for implementing a basic, read-only AI assistant are as follows:

(a) Build and configuration (one-off): approximately 6 to 12 full-time equivalent (FTE)-months (0.5 to 1.0 FTE-years); and

(b) Ongoing operations: approximately 0.5 to 1.5 FTE per year for monitoring, safeguards, prompt and tool governance and support, in addition to variable non-labor costs such as compute and inference, secure hosting, indexing and search infrastructure, and multilingual support, which would scale mainly with usage and scope.

VI. Modalities for the operationalization of the Mechanism that may need to be determined by the Conference of the Parties at its first meeting

131. The following parameters materially affect the architecture, functioning and cost of the Mechanism, and may therefore need to be determined by the Conference of the Parties at its first meeting, or addressed through interim guidance where feasible:

(a) Minimum required metadata fields for each type of record, including defined requirements for notifications on activities with respect to marine genetic resources and digital sequence information, mandatory attachments, and support for assisted or offline submission formats (e.g., Excel or Word formats);

(b) Definition and governance of confidentiality classifications (public, restricted and confidential), including criteria for classification and handling;

(c) User role taxonomy, verification rules and delegation mechanisms, including whether, and under what conditions, users from non-Parties may submit information;

(d) Safeguards for traditional knowledge of Indigenous Peoples and local communities, including minimum consent and provenance requirements, access request pathways, and whether traditional knowledge content may be hosted centrally or referenced through metadata with controlled access pointers;

(e) Policy on proximity-based alerts, if such a feature is provided, including applicable geographic triggers, permissible areas of interest and minimum geospatial data requirements;

(f) Interoperability priorities, including which external systems should be prioritized and which interoperability modes are authorized;

(g) Policy on nodes, if applicable, including whether nodes are endorsed, the applicable oversight and management model, and minimum security and operational standards for nodes;

(h) Language and translation policy, including which elements must be multilingual, expected service levels and the allocation of responsibilities; and

(i) Minimum cybersecurity baseline and audit requirements, including incident response expectations and data retention policies.

Part (c)

Draft roadmap for the phased operationalization of the Mechanism

I. Introduction

132. This part of the Study presents a draft roadmap for the phased operationalization of the Mechanism, reflecting a commonly used sequence for developing and launching complex information systems, including Phase 1 (Discovery), Phase 2 (Alpha), Phase 3 (Beta), and Phase 4 (Launch and maintenance). While different options or hybrid configurations may affect the emphasis, complexity or resource requirements within individual phases, the overall sequence is not altered.

133. The indicative resource requirements for Phases 1 through 4 cover the estimated costs for IT delivery experts, as well as for supporting infrastructure and external services, as discussed in Section VII of Part (c) below. Other resource requirements that may be required across all the options are addressed in paragraph 1, 2, and 6 of Appendix D.

II. Overall approach

134. The operationalization of the Mechanism can be approached as the launch of an ongoing service, rather than as a one-time technical build. In practical terms, this means that:

- (a) Each phase delivers clearly identifiable outputs that can be reviewed and agreed (for example, defined workflows, tested system functions or operational procedures);

- (b) Each phase includes structured opportunities for feedback from intended users and operators; and

- (c) Progression from one phase to the next is subject to a decision point confirming that minimum conditions are met before additional scope or complexity is introduced.

135. The phased approach reflects two core assumptions already set out in the Study:

- (a) The Mechanism will initially be operationalized as a minimum viable product and expanded over time; and

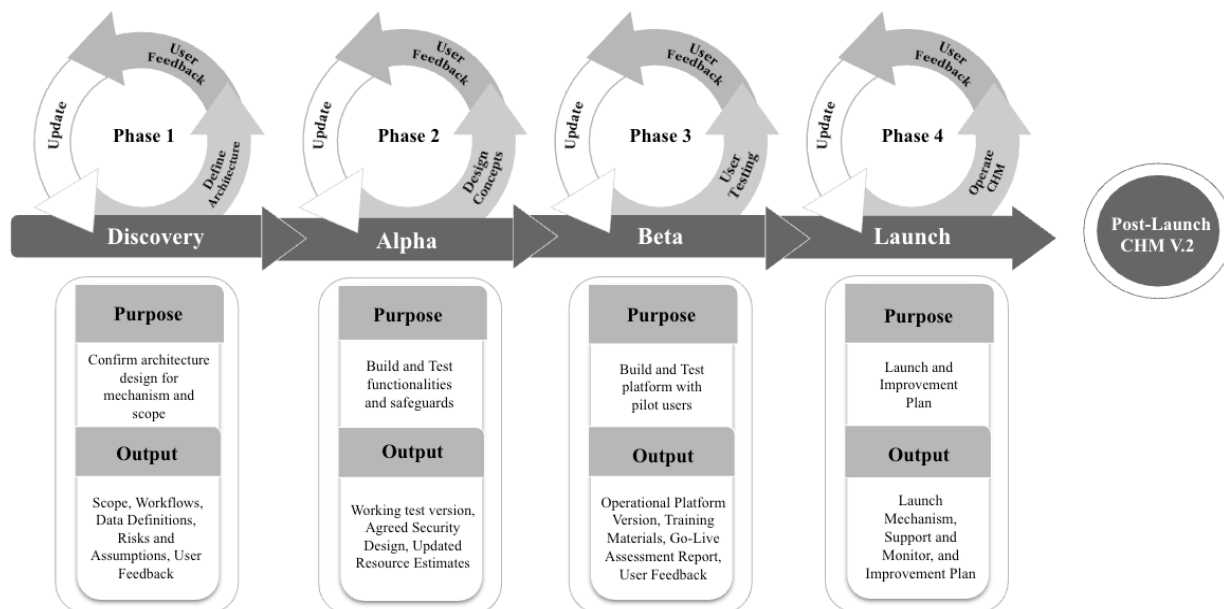
- (b) Some elements (e.g., deeper interoperability arrangements, the use of nodes, or certain confidentiality and traditional knowledge-related modalities) may need to be introduced progressively.

136. Each phase will generate a defined set of documents and operational evidence (e.g., an agreed scope baseline, test scripts and results, security assessment outputs and readiness checklists). Progression to the subsequent phase should be based on explicit acceptance criteria confirmed by the secretariat and, where relevant, any oversight arrangements mandated by the Conference of the Parties. These criteria may take into account relevant considerations identified in Part (a) of the Study (e.g., lessons learned from existing clearing-house mechanisms) and tiered interoperability with existing databases, information systems and clearing-house mechanisms explored in Appendix A. These acceptance criteria define the minimum conditions required to proceed to the next phase.

137. Figure 1 below illustrates the phased approach to the operationalization of the Mechanism.

Figure 1.

Illustration of the phased approach to the operationalization of the Mechanism



III. Phase 1 (Discovery)

138. The purpose of Phase 1 (Discovery) is to clearly define what the Mechanism must be able to do at the point of the initial operationalization. Activities in this phase will ensure that the requested scope is feasible, coherent and aligned with any decisions of the Conference of the Parties and other relevant factors, such as the availability of resources. Building on the Study, Phase 1 includes the confirmation of users and workflows, among others, including validation from an IT systems perspective that the identified workflows and constraints can be appropriately addressed through system design.

139. The key activities of Phase 1 are as follows:

(a) Confirming workflows: Confirm, step-by-step, how information will be submitted, reviewed, published, accessed and updated for each substantive element addressed by the Agreement, as well as for cross-cutting functions such as alerts, reporting and user support;

(b) Confirming practical constraints: Identify conditions that may affect the use of the Mechanism, such as low-connectivity environments, power outages and anticipated peaks in activity around deadlines or reporting cycles;

(c) Confirming any context-specific operating requirements: Depending on the arrangements for the functioning of the secretariat, including any institutional linkages with the United Nations, confirm any applicable institution-specific operating requirements, such as security and confidentiality requirements, that would need to be incorporated in the design;

(d) Confirming the functionalities for initial operationalization: Clearly distinguish functionalities that must be available at the time of initial operationalization from those that can be introduced later (e.g., data harvesting);

(e) Confirming policy-dependent parameters: Identify design choices that depend on decisions of the Conference of the Parties (e.g., confidentiality classifications, handling of traditional knowledge, any requirements for proximity-based alerts and expectations regarding nodes), and record such parameters explicitly as items requiring subsequent determination;

(f) Confirming roles and responsibilities of users and administrators: Clarify the roles and responsibilities of users and administrators during normal operation, including, for example, which steps are automated, which require secretariat review, and where national focal points or other actors are involved;

(g) Confirming standards for interoperability: Confirm the minimum set of common data fields and naming rules (e.g., record type, dates, submitting Party, geographic tags and stable identifiers) required to support the sharing, linking and retrieval of information across systems;

(h) Initial resource framing: Map the agreed scope against the workstreams identified in the Study (e.g., oversight, management and coordination, technical delivery, security and hosting, user support, content management, and interoperability management) to confirm that the proposed scope can be delivered within the available lead time and resources.

140. The outputs of Phase 1 include:

- (a) A clear definition of the scope for initial operationalization and for subsequent phases;
- (b) Descriptions of end-to-end workflows for users and operators;
- (c) A high-level description of information types, records and identifiers;
- (d) A list of key risks and assumptions; and
- (e) A plan for incorporating user feedback in subsequent phases.

141. User feedback is gathered through targeted consultations, such as structured interviews or short walk-throughs of proposed workflows, involving representative users, which may include secretariat staff, national focal points, as well as users from least developed countries, landlocked developing countries and small island developing States, relevant instruments, frameworks and bodies, civil society, the scientific community, the private sector and Indigenous Peoples and local communities.

142. Phase 1 is considered complete when the scope for initial operationalization is clearly defined and confirmed, end-to-end workflows are documented and validated with intended users, design choices depending on decisions of the Conference of the Parties are explicitly identified, roles and responsibilities of users and administrators are clarified, and the proposed scope is confirmed to be feasible in terms of resources, operating context, and lead time.

143. Phase 1 concludes with approval of the scope for initial operationalization, clarity on issues requiring decisions of the Conference of the Parties, and confidence that the most complex requirements can be tested in Phase 2 (Alpha), including through a working test version of key system functions based on agreed approaches to security, confidentiality and auditability, together with draft operating procedures for core tasks and updated resource estimates reflecting lessons learned.

IV. Phase 2 (Alpha)

144. The purpose of Phase 2 (Alpha) is to test whether the agreed design functions in practice, before committing to a full operational build. The focus of this phase is on validating functionality and safeguards.

145. The key activities of Phase 2 are as follows:

- (a) Testing core workflows: Implement and test the main sequences of actions required under the Agreement, such as submission of notifications, basic checks, publication, search and retrieval, and the generation of alerts;

(b) Testing higher-risk elements at an early stage: Test the generation and resolution of the “BBNJ” standardized batch identifiers; handling of confidential or proprietary information; treatment of traditional knowledge using metadata-only or controlled-access approaches; implementation of minimum proximity-based alerts, where applicable; and operation under low-bandwidth conditions;

(c) Confirming consistent presentation and terminology: Ensure that submission templates, record pages and user guidance are internally consistent and aligned with the terminology of the Agreement; and

(d) Preparing for operation: Begin drafting basic operating procedures for moderation, account management and user support at a level sufficient to assess operational feasibility.

146. The outputs of Phase 2 include:

- (a) A working test version of key system functions;
- (b) An agreed approach to security, confidentiality and auditability;
- (c) Draft operating procedures for core tasks; and
- (d) Updated resource estimates reflecting lessons learned.

147. User feedback is gathered by inviting users to carry out realistic tasks (e.g., submitting a notification, correcting errors, locating published information or subscribing to alerts) and to provide structured feedback, which is used to refine the design.

148. Phase 2 is considered complete when core operational workflows function end-to-end in a test environment, higher-risk elements are demonstrated in practice, security and audit approaches are workable, and remaining risks are understood and manageable.

149. Phase 2 concludes once the secretariat confirms that the acceptance criteria above have been met.

V. Phase 3 (Beta)

150. The purpose of Phase 3 (Beta) is to prepare the Mechanism for real-world use by a limited group of users, with an emphasis on operational readiness rather than design validation.

151. The key activities of Phase 3 are as follows:

- (a) Ensuring reliability and security: Test system performance, backups, logging, access controls and security measures at a level appropriate for operational use;
- (b) Confirming accessibility and multilingual support: Verify that the system can be used in low-connectivity environments, that navigation and core workflows function as intended, and that document handling performs reliably at expected volumes;
- (c) Preparing support arrangements: Finalize user guidance, training materials, and helpdesk processes, and confirm staffing and support assumptions; and
- (d) Running controlled pilots: Operate the Mechanism with representative users to confirm that it performs as expected in practice.

152. The outputs of Phase 3 include:

- (a) A near-final operational version of the Mechanism;
- (b) Training and support materials;
- (c) A go-live readiness assessment identifying any remaining issues; and
- (d) A consolidated record of user feedback and resulting adjustments.

153. Phase 3 is considered complete when the Mechanism operates in a near-final configuration, performs reliably under expected operating conditions, accessibility and multilingual support function as intended, user guidance and support arrangements are in place, and pilot use confirms readiness for initial launch with only minor documented issues remaining.

154. Phase 3 concludes once the Mechanism is secure, usable, supportable and fit for initial launch.

VI. Phase 4 (Launch and maintenance)

155. Phase 4 (launch and maintenance) covers the formal launch of the Mechanism and its transition into sustained, routine operation. The emphasis in this phase is on maintaining a reliable and secure system, monitoring real-world use, and introducing continuous improvements and any enhancements authorized by the Conference of the Parties beyond the initial minimum viable product. This phase treats the Mechanism as an evolving operational capability rather than a completed build.

156. The key activities of Phase 4 are as follows:

- (a) Launching and stabilizing operations: Bring the Mechanism into live use, with enhanced support during the initial period to resolve early issues, clarify workflows, and stabilize performance;

- (b) Operating and maintaining the Mechanism: Provide ongoing hosting, security, monitoring, backups, incident response, and routine maintenance to ensure continuity, integrity and availability over time;

- (c) Maintaining safeguards and operational controls: Operate confidentiality, audit, versioning, and publication controls in steady state, including agreed approaches for handling sensitive information and traditional knowledge;

- (d) Providing ongoing user support: Deliver sustained helpdesk and support, and use recurring issues to identify systemic usability or workflow improvements; and

- (e) Monitor usage, performance and data quality: Track submissions, alerts, access patterns, response times, error rates and data completeness to understand how the Mechanism functions in practice and to identify emerging friction, risks or opportunities for improvements.

157. The outputs of Phase 4 include:

- (a) A live, operational Mechanism in routine use;

- (b) Ongoing user support arrangements and updated user guidance;

- (c) Operational and usage information (e.g., submissions, alerts and performance indicators);

- (d) Incremental system enhancements and updated documentation; and

- (e) A forward-looking improvement plan aligned with decisions of the Conference of the Parties.

158. User feedback is collected continuously through helpdesk interactions, targeted user sessions and operational reporting, and is used to inform the prioritization of maintenance and improvement activities.

159. Phase 4 is considered to have commenced when the secretariat confirms that the Mechanism is live and accessible for intended users, operational responsibilities and support arrangements are in place, monitoring of usage and performance is active, safeguards operate in steady state, and a defined process exists for prioritizing and delivering ongoing improvements in line with decisions of the Conference of the Parties.

VII. Indicative cost estimates and lead time

160. Lead time refers to the elapsed time needed from a decision to proceed to the point of initial operationalization of the Mechanism. Lead time is highly sensitive to resourcing levels, delivery arrangements and implementation choices. The ranges set out below should therefore be read as order-of-magnitude indications rather than firm schedules.

161. The indicative cost estimates below reflect the estimated total cost to reach initial operationalization under each option. They are derived using a bottom-up, role-based approach. For each option, the delivery roles required to achieve initial operationalization were identified, together with the period for which those roles would be needed. Estimated levels of effort for each role (with some roles full-time and others time-limited or fractional) were multiplied by blended rates typical of international public-sector digital delivery. Non-staff costs are included to reflect infrastructure and tooling over the expected delivery period. Actual costs may vary depending on delivery arrangements, staffing choices, sequencing of work, and pricing of supporting services such as cloud hosting, software licenses and security testing. Other factors, such as the number of interface languages or refinements to alert features, may also affect costs at the margins.

162. The indicative lead time and cost estimates for Option A are as follows:

- (a) Indicative lead time: Approximately 9 to 12 months from the start of Phase 1.
- (b) Indicative cost estimates: Approximately USD 2 to 4 million.
 - (i) This range reflects the cost of a compact, multi-disciplinary delivery team responsible for end-to-end delivery, including product management, solution architecture, front-end and back-end development, user experience and accessibility design, quality assurance and testing, development operations and security engineering, and technical documentation and user guidance. Roles may be combined or provided on a part-time basis depending on the delivery arrangements and the final scope. The potential benefits and trade-offs associated with Option A, as discussed in Part (c) of the Study, may be considered.
 - (ii) The cost of IT delivery experts is expected to account for approximately USD1.4 to 3.0 million of the total.
 - (iii) Supporting infrastructure and external services are expected to account for approximately USD 0.6 to 1.0 million, covering cloud hosting for development, testing and production; managed storage and databases; monitoring and logging; identity and access management; code repositories and build pipelines; and time-limited security assurance activities such as vulnerability scanning and penetration testing. A basic level of multilingual interface support is assumed.

163. The indicative lead time and cost estimates for Option B are as follows:

- (a) Indicative lead time: Approximately 15 to 24 months from the start of Phase 1, depending in particular on how nodes are defined, introduced and resourced. Under Option B, the Mechanism is delivered as a central system managed by the secretariat, complemented by regional or subregional nodes that provide localized support, assisted workflows, or connectivity enhancements under agreed standards.
- (b) Indicative cost estimates: Approximately USD 3 to 6 million.
 - (i) As discussed in Part (c) of this Study, the higher cost range relative to Option A reflects the additional complexity associated with designing, coordinating and supporting a distributed delivery model. In addition to the central system described under Option A, this option requires additional effort for node-related design, integration, security assurance, coordination mechanisms, and supporting tooling to ensure that the central hub remains the single authoritative record.

- (ii) The cost of IT delivery experts under this option are expected to account for approximately USD 2.2 to 4.5 million of the total, reflecting higher overall staffing levels and longer involvement of architecture, security and interoperability roles.
- (iii) Supporting infrastructure and external services are expected to account for approximately USD 0.8 to 1.5 million, reflecting additional environments, integration tooling, security controls across multiple operating contexts, and higher coordination overhead.

164. For Option C, indicative lead time and cost estimates depend primarily on which existing platform is identified for retrofitting and its constraints, and can vary significantly depending on several factors, in particular the degree of customization required, including in relation to the level of security required, the types of user information to be collected, whether statistical or geospatial datasets are to be integrated (and to what extent visualization is required), the volume of confidential information, and the most suitable data and application architecture. Consultations with existing platforms undertaken for the purpose of the Study suggest that, for some platforms, lead time of up to three years and indicative cost estimates of approximately USD 5 to 6.3 million may be required. These indicative cost estimates are additional to any staffing resources that would be required for the secretariat for the management of the Mechanism

165. Illustrative hybrid configurations can vary widely, as they combine elements of Options A, B and C on a function-by-function basis. As a result, both the time required to reach initial operationalization and the total cost depend on the specific configuration selected, including whether node-supported workflows or retrofitted platform components are included. The indicative lead time and cost estimates are based on the illustrative hybrid example described in Appendix C of the Study, which applies node-supported approaches selectively for participation and consultation-intensive functions while retaining a central system for others.

- (a) Indicative lead time: Approximately 12 to 18 months from the start of Phase 1.
- (b) Indicative cost estimates: Approximately USD 2.5 to 5.5 million, derived using the same bottom-up, role-based costing approach applied to Options A and B. These figures apply only to this illustrative example; other hybrid configurations would entail different lead time and cost estimates.

VIII. Aspects following initial operationalization

166. Once the four phases have been completed, the Mechanism should be treated as an ongoing service. Any substantial new functionality or expansion beyond the initial scope of the initial operationalization should be addressed through a separate, subsequent project (e.g., a new version), with its own discovery, design and implementation process, as appropriate. The indicative timelines and resource estimates in this roadmap do not include such subsequent development, as such work would constitute a distinct effort requiring separate decisions and resourcing.

Part (d)

Draft workplan for the initial phase for the operationalization of the Mechanism

I. Introduction

167. Part (d) of the Study presents a draft workplan for the initial phase for the operationalization of the Mechanism. The objective of this part is to identify specific activities, outputs and resources requirements.

II. Pre-Phase 1

168. Prior to the beginning of Phase 1 (Discovery), the following activities would need to be undertaken:

- (a) Confirmation of option and key parameters: Choice of the overall option for the Mechanism (Options A, B or C,⁹ or a hybrid configuration¹⁰);
- (b) Clarification of the parameters identified in paragraph 131 of Part (b) of the Study; and
- (c) Acquisition of required capacity and allocation of required resources, including:
 - (i) Determination of the delivery arrangements for Phase 1, including whether the work will be undertaken through the creation of in-house capacity, engagement of consultants or individual contractors, using a vendor, or a combination thereof;
 - (ii) Identification and securing of the resources required, including through budgetary processes and resource mobilization, as appropriate; and
 - (iii) Hiring, contracting or procurement processes, as appropriate, to enable the selected delivery arrangements.

III. Phase 1 (Discovery) activities and outputs

169. The key activities of Phase 1 are as discussed in paragraph 139 of this Study. The key outputs are discussed in paragraph 140 of this Study.

IV. Indicative lead time and cost estimates for Phase 1

170. The indicative lead time range and cost estimates discussed below reflect the time and costs for discovery, design and scoping work to be undertaken by technical experts during Phase 1, but do not include time and costs that may be required for decision-making by the Conference of the Parties, the provision of guidance through any oversight arrangement, or engagement with and among Parties and relevant stakeholders. In addition, the time and costs that may be required for activities leading up to Phase 1 are also not included.

A. Indicative lead time range for Options A, B, and C

171. For Option A and Option B, Phase 1 serves the same purpose: to translate the selected option into an implementable and testable scope for initial operationalization. Phase 1 focuses on defining what must exist at launch, confirming workflows and responsibilities, identifying dependencies on decisions of the Conference of the Parties, and checking that the proposed scope can realistically be delivered within the assumed resourcing and lead time. The activities and outputs of Phase 1 are substantively the same across both options; what differs is the complexity and effort required. These indicative differences should be interpreted alongside risk reduction and feasibility benefits identified through Phase 1 assessments, which may reduce rework in later phases.

172. Under Option A, Phase 1 is more straightforward because all functions are handled centrally. There are no nodes to define, no inter-entity coordination rules to establish, and fewer complexities to resolve. As a result, the indicative estimated lead time for Phase 1 for Option A is approximately four to six weeks, corresponding to roughly 60 to

⁹ Option C would require a decision on the existing platform that would be retrofitted and on the oversight and management arrangements (a central system only or a central system plus nodes).

¹⁰ A hybrid configuration would require a decision on the combination of the functional elements.

100 working days across a compact, multi-disciplinary core team using the same assumptions applied elsewhere in this roadmap.

173. Under Option B, Phase 1 follows the same steps but requires additional definition work up front. Such additional work includes clarifying what nodes do and do not do, how information flows between nodes and the central system, and how security, confidentiality, and safeguards for traditional knowledge are applied consistently across the network. Because of the consequent added coordination and design effort, the indicative estimated lead time of Phase 1 for Option B is approximately six to eight weeks, corresponding to roughly 90 to 150 working days across the core team under the same assumptions.

174. These indicative timelines are illustrative only. Actual duration and effort will depend on the scope developed in Phase 1, and delivery arrangements. The estimates are intended to provide a consistent basis for comparison across options.

175. For Option C, indicative estimated lead time for Phase 1 is not available given the uncertainties discussed in paragraph 164 of Part (c) of the Study.

B. Indicative lead time for the illustrative example of a hybrid configuration (see Appendix C)

176. The illustrative example of a hybrid configuration, presented in Appendix C of the Study, demonstrates how a combination of elements from Options A, B and C may be chosen for the operationalization of the Mechanism, with more detailed discussions in connection with its functions. The IT delivery experts would then translate the information to determine the IT architecture and design required to build the Mechanism.

177. The following paragraphs use the illustrative hybrid example provided in Appendix C to demonstrate how the Phase 1 steps discussed above may be applied in practice, including the indicative roles involved, the level of effort required, the activities undertaken, and the outputs produced.

178. For the illustrative hybrid example, Phase 1 is assumed to take approximately six weeks. During that time, functions will be selected from the hybrid configuration and translated into an implementable and testable scope for initial operationalization. Phase 1 is intended to produce a validated understanding of what must be delivered at operationalization, what may be deferred, which depends on decisions of the Conference of the Parties, and what is feasible within the assumed resourcing and lead time. This, in turn, enables Phase 2 (Alpha) to prioritize building key components and test them at an early stage.

179. For the illustrative hybrid example, the following indicative lead time may be required:

(a) Product manager (0.8 to 1.0 full-time equivalent (FTE), approximately 24 to 30 working days): Overall responsibility for Phase 1 delivery, scope control, prioritization, timelines, and coordination across technical workstreams; primary liaison with the secretariat and stakeholders, as appropriate.

(b) Technical lead and solution architect (0.5 to 0.8 FTE, approximately 15 to 24 working days): Defines system security rules for the hybrid model, validates feasibility of reuse and node-supported participation, and record preservation.

(c) User experience and service designer (0.3 to 0.5 FTE, approximately 9 to 15 working days): Designs user journeys, submission flows, alerts, and accessibility features for central workflows, node-supported participation, and low-connectivity use.

(d) Software engineers (front-end and back-end) (0.4 to 0.8 FTE combined, approximately 12 to 24 working days): Supports Phase 1 through technical consultations

and feasibility checks (e.g., retrofit platform fit for user management and search, alert distribution patterns, basic interoperability assumptions).

(e) Data and systems analyst (0.3 to 0.5 FTE, approximately 9 to 15 working days): Defines record types, database setup, data pipelines, and validation logic needed.

(f) Documentation and requirements specialist (0.2 to 0.3 FTE, approximately 6 to 9 working days): Translates requirements into engineering specifications, workflow descriptions, risks and assumptions, and decision logs.

(g) Development operations and cloud operations specialist (0.1 to 0.2 FTE, fractional, approximately 3 to 6 working days): Advises on hosting, environments, and operational constraints

(h) Security specialist (0.1 to 0.2 FTE fractional and on-demand, approximately 3 to 6 working days): Reviews confidentiality handling, safeguards for traditional knowledge, access control and auditability.

C. Indicative cost estimates for Phase 1

180. The indicative cost estimates outlined below are based on the assumption that, while Phase 1 may not be resource-intensive in terms of the working days or headcount for IT delivery experts, common practice typically requires the front-loading of senior technical expertise and coordination effort at the outset of Phase 1 in order to establish scope, reduce delivery risk and mobilize the work. It is further noted that the cost of Phase 1 may be at the higher end of the indicative range where the initial scope of the IT delivery work is not sufficiently defined. For example, where only a broad guidance, such as reference to Option B, is provided without specification of detailed functionalities, additional resources may be required during Phase 1 to determine detailed functionalities. On this basis and for the purpose of the Study, the indicative costs for Phase 1 are assumed to represent approximately one quarter of the total indicative cost estimates for the initial operationalization of the Mechanism (see Part (c) of the Study), noting that Phase 1 is one of four phases.

181. Accordingly, the indicative cost ranges for Phase 1 are as follows, derived from the total indicative cost estimates outlined in Part (c) of the Study:

(a) For Option A: approximately USD 0.5 to 1 million;

(b) For Option B: approximately USD 0.75 to 1.5 million; and

(c) For the illustrative example of a hybrid configuration: approximately USD 0.6 to 1.25 million.

182. For Option C, an indicative cost estimate for Phase 1 is not available due to the uncertainties discussed in Part (c), paragraph 164, of the Study.

183. In addition to the indicative cost estimates outlined above, which cover the estimated costs for IT delivery experts, as well as for supporting infrastructure and external services, there may be additional costs, as outlined in Appendix D of the Study.

Appendix A

Indicative and non-exhaustive external source integration matrix

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Appendix B

Cost assessment approach

1. The resource requirements for the operationalization of the Mechanism will depend on various factors, which may include decisions by the Conference of the Parties on scope, workflows, oversight and management arrangements, procurement choices, and final service-level decisions. In the absence of clarity on these factors, estimated resource requirements are primarily expressed as estimated staffing requirements rather than fixed monetary figures. In other words, the estimated costs answer the question “what roles, how many full-time equivalent staff, performing which functions, and for what duration” before translating resource requirements into budget lines under different contracting and staffing models.

2. The cost profile of each option is expressed as:

(a) Estimated headcount (Full-Time Equivalent, or FTE) required to build and to operate the Mechanism;

(b) Indicative role mix and workstreams (i.e., team composition);

(c) Key non-labor cost drivers (e.g., hosting, cybersecurity testing, translation approach, specialist tooling); and

(d) Cost incidence, indicating the distribution of costs among the secretariat, Parties and external partners, recognizing that some options may involve national, regional or subregional actors, as is the case for some platforms discussed in Part (a) of the Study.

What is a “headcount” (FTE)?

3. For the purpose of the Study:

(a) 1.0 FTE corresponds to one person working full-time for one year on Mechanism-related functions (e.g., staff member, volunteer, consultant, or equivalent vendor capacity).

(b) Headcount estimates are presented as ranges to reflect uncertainties related to the modalities for the operation of the Mechanism.

(c) Functional roles may be combined in various ways, such as product management combined with policy liaison, Development-Security-Operations (DevSecOps)¹¹ combined with platform engineer, or dedicated security officer combined with dedicated training lead.

Two complementary headcount approaches

4. Each option presents resource requirements using two complementary approaches, applied consistently across options:

(a) Secretariat-resourced FTE (central delivery capacity). Effort required within the secretariat, or directly contracted by it, to build and operate the Mechanism. This includes service management, engineering, DevSecOps and cybersecurity, information management, user support, and interoperability governance.

(b) Distributed “ecosystem” FTE (incurred outside the secretariat). Effort borne by Parties, and, where applicable, by regional or subregional entities and other external partners to enable the Mechanism to function in practice. This may include, for example,

¹¹ Development-Security-Operations (DevSecOps) refers to the combined functions of software development, system operations, and ongoing security tasks such as hosting, monitoring, patching, backups, vulnerability management, and incident response.

focal point validation and approvals; preparation and quality assurance of submissions; inter-agency coordination; partner onboarding for interoperability; and localized user support in more distributed models.

5. These approaches do not presume a single funding model. They are intended to make explicit the workload distribution, including where capacity may be provided in-kind, supported through external assistance, or financed pursuant to decisions of the Conference of the Parties (e.g. through assessed contributions), even where functions are carried out by entities outside of the secretariat.

How headcount is estimated for each option

6. Each option is assessed across two time horizons:

(a) Build and configure (one-off): Expressed as average FTE over the delivery period (e.g., an initial operationalization delivery window), plus any short-term surge needs (e.g., penetration testing, accessibility conformance reviews, and migration or import tasks); and

(b) Operations (recurring): Expressed as annual steady-state FTE required to operate, support, secure and incrementally improve the Mechanism after initial operationalization.

7. To ensure comparability across options, staffing estimates are grouped into consistent workstreams, which also help identify where functions may realistically be combined:

(a) Product and governance (e.g., product owner and service manager, policy liaison, release planning, reporting to the Conference of the Parties and subsidiary bodies);

(b) Engineering (e.g., solution architecture, application development, testing and quality assurance and documentation);

(c) DevSecOps and cybersecurity (e.g., hosting, monitoring, patching, backups, vulnerability management and incident response);

(d) User support and adoption (e.g., helpdesk; onboarding; training; moderation and triage; facilitation support for workflows such as environmental impact assessment consultations and needs and priorities assessments in terms of capacity-building and the transfer of marine technology);

(e) Content and information management (e.g., record quality checks, metadata management, redaction workflow support and publication quality assurance); and

(f) Interoperability and standards lifecycle (e.g., partner onboarding, interface and memoranda of understanding management; metadata and vocabulary maintenance; application programming interface change management).

Cost buckets

8. The staffing and non-labor drivers for each option are described using four cost buckets which range from one-off to recurring, applied consistently across options:

(a) Build and configure (one-off): architecture and design, development and configuration, templates and workflows setup, security baseline, initial integrations, content setup, testing and go-live readiness:

(b) Operations (recurring): hosting, DevSecOps, maintenance, monitoring, cybersecurity, backups, incident response and routine upgrades; and

(c) Human support (recurring): helpdesk support, multilingual support approach, training and onboarding, moderation and curation, and facilitation support for key workflows.

(d) Interoperability lifecycle (recurring and periodic): partner onboarding, memoranda of understanding management, application programming interface change management, and standards and vocabulary maintenance.

9. User support and efforts to ensure correct and consistent use of the Mechanism are treated as core costs for the operation of the Mechanism, not optional add-ons.

Baseline assumptions for comparability

10. Given the uncertainties related to the modalities for the operation of the Mechanism, FTE estimates are provided as ranges and are accompanied by clear statements identifying the assumptions that most affect staffing requirements. For comparability across options, the following assumptions are applied:

(a) Interoperability at the initial operationalization: assumed to rely primarily on link mode (i.e., authoritative links with standardized metadata). More complex interoperability (i.e., metadata harvesting and application programming interface federation) is treated as a tiered enhancement, which would require partner agreements, security arrangements, and ongoing change management.

(b) Multilingualism: multilingual navigation and core workflows are treated as a platform requirement, with a pragmatic approach that may include automation or artificial intelligence-enabled translation for selected content types. It is not assumed that the secretariat will provide human translation for all submitted documents. Any human translation requirements mandated by the Conference of the Parties are identified as cost drivers with staffing implications.

(c) Helpdesk: assumed to be business-hours support via ticketing and email supplemented by scheduled onboarding and support clinics, with escalation pathways for deadline-sensitive requests. Expanded coverage models (e.g., regionalized desks, phone or application-based messaging channels, and extended hours) are treated as scalable enhancements with corresponding FTE impacts.

(d) Volumes and throughput: submission and user volumes are not yet known. Accordingly, recurring support and moderation are estimated using volume bands and described in terms of, first, fixed baseline capacity, and second, variable capacity that scales with throughput.

(e) Confidential or restricted content share: the Mechanism is assumed to be primarily open access, with some submissions requiring confidential or restricted handling and safeguards for traditional knowledge. If the share of confidential or restricted content is higher than assumed in practice, moderation, redaction, audit and governance workload scales non-linearly and is treated as a material cost driver.

Translating headcount into budget

11. A monetary budget can be derived by applying loaded cost rates to each FTE category (e.g., United Nations staff grades, United Nations volunteers, consultants or vendor service contracts), together with non-labor line items such as hosting, licenses, security testing, translation accessibility services. However, these rates and overhead assumptions vary substantially depending on the following factors:

(a) employment model (e.g., staff member, consultants or vendor-managed service);

(b) level of experience of the given individual (estimates in this Study use average rates);

(c) hosting approach;

(d) multilingual policy;

(e) required support hours and training expectations;

- (f) volume of records; and
- (g) the number and complexity of integrations.

12. For this reason, the cost assessment focuses on headcount ranges, key non-labor cost drivers and cost incidence discussed in paragraph 2 above to provide a basis for budgeting.

Appendix C

Illustrative examples of functionalities for Options A through C

1. Table A below provides an illustrative example of how the functions of the Mechanism could be implemented under each of the three options described in the Study (Options A, B, and C).
2. Regarding cost considerations, differences between options are driven primarily by choices concerning oversight and management arrangements and the delivery modalities. Detailed resource requirements are therefore addressed separately through the accompanying analysis of full-time equivalent (FTE) and related narrative discussion. The table below does not prescribe a preferred configuration, nor should it be interpreted as a menu of independent features. Several functions are interdependent, and choices in one area may have implications for oversight and management arrangements, safeguards, confidentiality, or the integrity of the authoritative record in others.

Table A.

Illustrative example of how the functions of the Mechanism could be implemented under Options A to C

<i>Functions</i>	<i>Option A (central system only)</i>	<i>Option B (central system + nodes)</i>	<i>Option C (retrofitting existing platform)</i>
User management (roles, verification, audit)	Approach: The central system manages user accounts, assigns roles, verifies authorized users, and records user actions for oversight and review. Pros: A single set of user roles and access rules, defined and applied centrally. Cons: Most user registration and verification tasks are handled by the secretariat through the central system, and limit hands-on support at regional or subregional level.	Not a selectable option for this function: User management remains unchanged as Option A and is handled entirely by the central system. Regional or subregional nodes would provide user support or training without altering the IT design or responsibilities of the central system.	Approach: User management is handled through an existing platform's account and access system, with BBNJ user roles and verification requirements implemented within that platform's existing structure. Pros: Faster to implement if the platform already supports user approval, role assignment, and access records. Cons: The structure of user roles and access rules is limited by what the platform supports, and changes may require platform-specific updates rather than simple adjustments by the secretariat.
Submission and publishing (forms, uploads, versioning)	Approach: The central system accepts submissions and publishes official records. Submission forms, document uploads, validation, version control, and publication status are handled in one place. Pros: Reduces ambiguity on what constitutes the official record and, therefore, the risk of inconsistent or conflicting submissions. Cons: All submission and publishing work is handled through the central system, placing operational responsibility on the secretariat.	Approach: Submissions are prepared or checked with support from regional or subregional nodes before being sent to the central system for publication. This staging may be carried out using local systems, third-party tools, or manual processes. Pros: Reduces incomplete or incorrect submissions reaching the central system and allows support to be provided closer to users. Cons: Introduces additional coordination and workflow steps, which require clear rules and training to avoid delay or inconsistency.	Approach: Submission and publishing use an existing platform's workflow and document-handling tools, with BBNJ-specific forms and defined publication steps configured within that platform. Pros: Faster to implement if the platform already supports structured submissions, versioning, and publication workflows. Cons: Submission and publishing behavior is limited by what the platform can support, and additional configuration or customization may be needed to meet BBNJ-specific requirements.
Search and retrieval (indexing, filters, record pages)	Approach: The central system provides search and browsing for all records. Indexing, filtering, and record views are handled directly within the central system. Pros: The way a user searches is the same way for all records, reducing confusion about what users can see and how they can find it. Cons: All search and browsing capacity must be provided by the central system, and more advanced features may require additional development over time.	Approach: The central system controls which records can be searched and how they are structured, while search access may be provided through regional, language-specific, or pillar-specific interfaces. Pros: Users can find records through regional or pillar-specific interfaces users already use, without changing which system controls access and visibility. Cons: Clear rules are needed to keep external search interfaces in sync with the central system, so users do not see outdated or partial records.	Approach: Search and record pages use an existing platform's search and indexing features. Pros: Faster to implement if search and filtering are already available. Cons: Search features are limited to what the platform supports, which may restrict filters, sorting, or geospatial views, and adding new capabilities may require significant customization or may not be possible.
Notifications and alerts (subscriptions, digests, proximity)	Approach: The central system controls when alerts are triggered and sends notifications directly. Alerts are issued for defined events such as submissions, updates, deadlines, and where applicable, based on agreed geographic criteria. Pros: Alert timing and content are controlled in one place, reducing confusion about when and why notifications are sent. Cons: All alert delivery is handled by the central system, with no regional or node-supported dissemination.	Approach: The central system defines when alerts are triggered and may deliver alerts directly. Alerts may also be distributed through nodes or other external systems operating under central system rules, which relay the same alerts to users without changing their timing or content. Pros: Allows alerts to reach users through regional or pillar-specific arrangements, rather than relying only on central delivery. Cons: Clear rules are needed to ensure alerts distributed through other systems remain timely and consistent.	Approach: Use an existing platform's alerting capability to implement the alert rules, triggers, and notification sending within that platform. Pros: Faster implementation if alerting and geospatial features already exist which meet the BBNJ Agreement's requirements Cons: Alert rules and triggers may be constrained by what the platform supports, and changes may require customization or may not be possible.

<i>Functions</i>	<i>Option A (central system only)</i>	<i>Option B (central system + nodes)</i>	<i>Option C (retrofitting existing platform)</i>
Confidentiality controls (classification, redaction)	Approach: The central system requires submitters to identify whether information is public, restricted, or confidential at the time of submission. It applies access restrictions accordingly, and supports redaction for public release. Pros: Confidentiality rules and redaction practices can be defined and changed directly by the secretariat, without relying on the constraints or upgrade cycles of an external platform. Cons: All review and redaction are handled by the central system, which may limit the ability to reflect local or community context when handling sensitive information.	Not a selectable option for this function: Confidentiality controls cannot be meaningfully distributed. Decisions on access restrictions, redaction, and handling of sensitive information must remain centralized to avoid inconsistent treatment or loss of control. Nodes may assist users in preparing submissions, but they would not apply confidentiality decisions or control access to sensitive information.	Approach: Confidentiality handling uses an existing platform’s access controls, permission settings, and audit features, with requirements specific to the BBNJ Agreement implemented within that platform. Pros: Faster to implement if the platform already supports restricted access, redaction, and access logging. Cons: Confidentiality handling is limited by what the platform supports, and adapting it to needs specific to the BBNJ Agreement may require customization or may not be possible in all cases.
Low-connectivity accessibility	Approach: The central system is designed to be usable with slow or unreliable internet connections, using simple pages and processes. All users interact directly with the central system. Pros: Keeps access and use of the Mechanism simple, with one system to maintain. Cons: Users still need a live connection to the central system to submit or access information, with no offline or deferred submission option when connectivity is unavailable.	Approach: The central system remains responsible for records, while nodes help users work with limited connectivity by allowing preparation and submission of information when a stable connection is available. Pros: Makes it easier for users with poor connectivity to participate without needing continuous access to the central system. Cons: Adds coordination work to ensure information prepared through nodes is transferred correctly and on time.	Approach: Use an existing platform’s low-connectivity features such as offline use or delayed submission as the primary means of supporting users with limited or unreliable internet access. Pros: Faster to implement if the platform already supports use with limited or unreliable connectivity. Cons: The level of low-connectivity support is limited to what the platform provides and may not align fully with user needs.
Notifications on activities with respect to marine genetic resources and digital sequence information and “BBNJ” standardized identifier	Approach: The central system receives notifications, checks that required information is provided, generate the “BBNJ” standardized batch identifier, and makes the identifier publicly retrievable together with the official record relating to activities with respect to marine genetic resources and digital sequence information. Pros: Ensures a single, global unique identifier is issued and used consistently across all notifications and updates on activities with respect to marine genetic resources and digital sequence information. Cons: The issuance process must be reliable and continuously available, and may require dedicated staffing, especially if the process is not fully automated.	Not a selectable option for this function: The generation of the “BBNJ” standardized batch identifier must be carried out by a single global system and cannot be distributed. Nodes may help users prepare notifications or reduce errors before submission, but it would not be advisable that they generate identifiers or control records relating to activities with respect to marine genetic resources and digital sequence information in the interest of the integrity of the process.	Not a selectable option for this function: Reusing an existing platform does not remove the need to build a single process for generating the “BBNJ” standardized batch identifier, identifier rules, and public lookup function. These elements must still be implemented specifically for the BBNJ Agreement, regardless of the underlying platform.

<i>Functions</i>	<i>Option A (central system only)</i>	<i>Option B (central system + nodes)</i>	<i>Option C (retrofitting existing platform)</i>
Workflows relating to area-based management tools, including marine protected areas	Approach: The central system publishes proposals for the establishment of area-based management tools, including marine protected areas, consultation materials, decisions, and updates. Record pages, version history, and key dates are managed in one place. Pros: All information and deadlines relating to area-based management tools, including marine protected areas, are published through a single official record, making it clear what is current and authoritative. Cons: All participation and consultation activity must occur through the central system, which may face challenges for effective engagement in contexts affected by connectivity or capacity constraints.	Approach: The central system publishes and maintains the official ABMT record. Nodes or other approved tools may be used to support participation in consultations, with inputs collected through those channels incorporated into the central record. Pros: Enables participation through tools and arrangements that stakeholders already use, while keeping one official record relating to area-based management tools, including marine protected areas. Cons: Clear procedures are needed to ensure inputs collected outside the central system are complete and accurately reflected in the official record.	Approach: Consultations on area-based management tools, including marine protected areas, and document handling use an existing platform's consultation and record-management features, adapted to reflect the stages and publication requirements for area-based management tools, including marine protected areas. Pros: Quicker to implement if consultation workflows and version tracking are already available. Cons: The platform's workflows may not fully match processes relating to area-based management tools, including marine protected areas, under the BBNJ Agreement and may require customization.
Workflows relating to environmental impact assessments	Approach: The central system manages the full environmental impact assessment process, including submissions, consultations, updates, and decisions. All information relating to environmental impact assessments is maintained in a single record with a single timeline. Pros: Keeps environmental impact assessment steps, deadlines, and decisions clearly ordered in one place, making the official process easy to follow. Cons: All consultation management and document handling are concentrated in the central system, which can be difficult to manage for long, document-heavy, or high-volume environmental impact assessment processes.	Approach: The central system maintains the official environmental impact assessment record and timeline. Nodes may support participation in consultations using their own tools, with inputs collected through those channels incorporated into the central record. Pros: Allows consultation support to be provided closer to participants, which can improve engagement and the completeness of submissions. Cons: Clear procedures are needed to ensure inputs gathered outside the central system are complete and accurately reflected in the official environmental impact assessment record.	Approach: environmental impact assessment workflows use an existing platform's consultation and document-handling features, adapted to reflect the required EIA steps and publication requirements. Pros: Quicker to implement if consultation and document-handling workflows already exist. Cons: The platform's workflows may not fully match the environmental impact assessment processes under the BBNJ Agreement and may require customization.
Matching of capacity-building needs with the support available and with providers for the transfer of marine technology	Approach: The central system hosts a single register of capacity-building needs and offers, matches needs with offers, and produces related summaries and reports. Pros: All needs and offers are listed and matched in one place using the same categories, making results easy to compare and report. Cons: How needs and offers are described and matched is set centrally, which can limit tailoring to regional or sector-specific contexts.	Approach: The central system maintains the official register of needs and offers. Nodes may use their own tools to help gather and present information, with the results entered into the central register. Pros: Information can be collected and support provided at the regional and local levels, which can improve the clarity and usefulness of needs and offers. Cons: Clear procedures are needed to ensure information collected through nodes is complete and entered consistently in the central register.	Approach: Capacity-building and transfer of marine technology needs and offers are managed using an existing platform's needs-and-offers features, adapted to reflect the Agreement's categories and reporting requirements. Pros: Quicker to implement if the platform already supports collecting needs, listing offers, and matching them. Cons: How needs and offers can be described, matched, and reported may be limited by the platform. Customization may be required and may not be possible in all cases.

<i>Functions</i>	<i>Option A (central system only)</i>	<i>Option B (central system + nodes)</i>	<i>Option C (retrofitting existing platform)</i>
Interoperability	Approach: The central system connects to external systems using links only. For each external database, the central system provides a reference and directs users to view information at the source. Pros: Lowest dependency and fastest to implement across many external systems. Cons: Information remains distributed across systems, and users must move between sources to view or compare data.	Approach: For selected external databases, the central system goes beyond links and automatically retrieves or exchanges information using agreed data feeds or system-to-system connections. This choice is made on a database-by-database basis. Pros: Allows tighter integration and reduces manual effort for users where external systems support automated exchange. Cons: Requires agreements, technical alignment, and ongoing coordination for each external database that uses this approach.	Not a selectable option for this function: Interoperability choices are made per external database (link-only versus automated exchange). Reusing an existing platform cannot remove or replace these decisions and does not constitute a separate interoperability design.
Traditional knowledge	Approach: The central system sets the rules for when traditional knowledge may be shared and handles all requests to view it. By default, records indicate the existence of the knowledge without revealing its content. Pros: Rules for sharing traditional knowledge are applied in one place, reducing the risk of inconsistent or accidental sharing. Cons: Handling all requests through the central system can be distant from communities and may require extra effort to reflect local language and cultural practices.	Approach: The central system sets the rules for when traditional knowledge may be shared. Nodes support communication with communities, help explain the process, and assist with requests. Sensitive information may remain with node-supported arrangements, while the central system records whether consent exists. Pros: Support can take place closer to communities while the rules for sharing remain the same everywhere. Cons: Clear boundaries are needed so community support does not result in different sharing rules in different places.	Approach: Traditional knowledge requests and access are handled using an existing platform's built-in request and access features, adapted to follow the BBNJ Agreement's requirements. Pros: Faster to implement if the platform already supports access requests and restricted viewing. Cons: Some platform features may need adjustment to reflect consent requirements and handling practices under the BBNJ Agreement.

3. Below is an illustrative example of a hybrid configuration using the functions discussed in table A above to demonstrate how elements of Options A to C can be combined to meet the baseline requirements:

(a) User management: Option C (retrofit an existing platform for login, role assignment and access records if it can implement roles and the approval steps of the secretariat required under the BBNJ Agreement);

(b) Submission and publishing: Option A (keep submission and published records in the central system to ensure a single official record of what is submitted, changed and published);

(c) Search and retrieval: Option C (retrofit an existing platform for search and record pages, provided it can support the needed record structure, filters and, if required, geospatial views);

(d) Notifications and alerts: Option B (keep alert triggers defined centrally, but allow nodes, and other external systems operating under central system rules, to distribute the same alerts to users);

(e) Confidentiality controls: Option A (keep access restriction and redaction handled through the central system to ensure that the handling of sensitive and/or confidential information does not vary by location);

(f) Low-connectivity accessibility: Option B (use nodes to support offline preparation and deferred submission once a stable connection is available, without changing where records are published and maintained);

(g) Notifications on activities with respect to marine genetic resources and digital sequence information and the “BBNJ” standardized batch identifier: Option A (keep the generation of identifier and the public lookup function in the central system, as a single central issuer is required, while nodes can help reduce errors before submission); (h) Workflows relating to area-based management tools, including marine protected areas: Option B (keep the official record and key dates in the central system, while using nodes to support consultation participation and ensure inputs are incorporated into the central record);

(i) Workflows relating to environmental impact assessments: Option B (keep the official record and timeline in the central system, while using nodes to support consultation participation and improve the completeness of inputs);

(j) Matching of capacity-building needs with the support available and with providers for the transfer of marine technology: Option B (keep the official register of needs, opportunities and providers in the central system, while using nodes to help Parties and stakeholders describe needs, opportunities and providers clearly to support effective matching);

(k) Interoperability: Option B (determine interoperability on a platform-by-platform basis, with link-only approaches applying more broadly, while automated exchange being limited to a small set of priority external platforms where it is feasible and justified by the coordination efforts required); and

(l). Traditional knowledge: Option B (keep the rules for when traditional knowledge may be shared defined centrally, while using nodes for communication and facilitation closer to communities).

4. An illustrative hybrid configuration such as the example in paragraph 3 above could entail an indicative level of effort in the range of approximately 3 to 6 FTE over a year for the IT architectural design and build, with additional effort by the secretariat on coordination where nodes are deployed and separately staffed and coordination between the IT team and relevant stakeholders.

Appendix D

Examples of costs associated with the operationalization of the Mechanism

1. Certain human resources may be required within the secretariat to support the operationalization of the Mechanism. It is noted that: first, some roles may be combined; second, some roles may have responsibilities beyond the Mechanism; and third, some roles may be supported through nodes where nodes are used. Indicative roles that may be required, on either a full-time or part-time basis, include:

(a) Programme manager: Overall management of Phase 1 delivery, including prioritization and sequencing of work, coordination across policy, technical and operational streams, supporting any oversight arrangements, budget planning and management, and reporting to the Conference of the Parties and subsidiary bodies;

(b) Interoperability officer: Coordination with relevant instruments, frameworks and bodies, other clearing-house mechanisms and information-sharing platforms, repositories and databases, including early alignment on interfaces, identifiers and information-sharing arrangements;

(c) Legal officer: Management of confidentiality classifications, data-sharing arrangements, and legal and procedural compliance, including support related to external services and contractual arrangements;

(d) Traditional knowledge officer: Management of safeguards for traditional knowledge, including consent metadata, and access conditions, and coordination with Indigenous Peoples and local communities on processes;

(e) Accessibility officer: Ensuring system design and delivery reflect connectivity constraints and accessibility needs, particularly for least developed countries, landlocked developing countries and small island developing States, including providing guidance to the technical teams and external service providers;

(f) Capacity-building and transfer of marine technology officer: Design of initial workflows, templates and guidance, and preparation for needs assessment and matching functions;

(g) Information systems officer: Technical management of the Mechanism during build and early operations, including coordination with external IT service providers, security and hosting arrangements, system availability, configuration management, and go-live readiness;

(h) Administrative officer: Administrative and operational support, including processing of administrative actions, contract administration, scheduling, travel logistics, and coordination of documentation; and

(i) User support and data operations officer: Helpdesk and onboarding support, first-line user assistance, routine verification and quality checks on submissions, and operational support during initial use of the Mechanism.

2. Non-post administrative resources may include costs associated with office space, furniture and IT equipment (e.g., laptop or desktop computers).

3. Complex IT projects such as the Mechanism typically require a team of IT professionals with a range of complementary skills. While the titles may vary and the functions may be combined, the following roles reflect the types of expertise that are typically required:

(a) Product manager: Overall responsibility for delivery, scope control, prioritization, timelines, and coordination between technical workstreams;

- (b) Technical lead and Solution architect: Overall system design, technology choices, security baseline, integration approach and ensuring that the architecture supports agreed requirements;
 - (c) User experience and service designer: Design of user journeys, submission flows, alerts, and accessibility-by-design features, informed by representative user needs rather than detailed policy ownership;
 - (d) Software engineers (front-end and back-end): Implementation of core workflows, submission forms, search and retrieval, alerts, and basic interoperability features required for initial operationalization;
 - (e) Data and systems analyst: Definition of data models, metadata fields, identifiers and validation logic required for consistent records and future interoperability;
 - (f) Documentation and requirements specialist: Preparation of functional specifications, user guidance and technical documentation to support implementation, testing and handover;
 - (g) Development operations and cloud operations specialist: Setup and operation of hosting environments, deployment pipelines, monitoring, backups and basic operational resilience for go-live; and
 - (h) Security specialist: Security review of architecture and implementation, including access control, data handling, and basic penetration testing.
4. The expertise referred to in paragraph 3 above may be sourced through different arrangements, including through engaging consultants or individual contractors, or through procurement of a bundled external service. A bundled external service may, in some cases, reduce the administrative burden on the secretariat, including in relation to recruitment, contracting, payments and supervision. Under such an arrangement, responsibility for coordination within the IT team would typically rest with the service provider, not the secretariat. At the same time, engaging consultants or individual contractors may, in some cases, offer greater flexibility and continuity.
5. Other IT-related costs associated with supporting infrastructure and external services may include cloud hosting and storage, development and test environments, security services, software licenses or managed services, backup and monitoring, and other infrastructure needed for system operation.
6. Other relevant costs may include:
- (a) Translation costs: the range of costs would depend on the extent to which the Mechanism and the documents housed in the Mechanism will be multilingual and their volume.
 - (b) Costs related to oversight arrangements: e.g., intergovernmental or expert meetings or workshops, either with or without document translation and interpretation.
 - (c) Engagement costs: e.g., communication with stakeholders; outreach to raise awareness, which may include workshops, either with or without document translation and interpretation, webinars, and information materials.